dSPG: A New Discriminant Superpixel Graph Regularizer and Convolutional Network for Hyperspectral Image Classification

Long Yu, *Graduate Student Member, IEEE*, Jun Li, *Fellow, IEEE*, Lin He, *Member, IEEE*, Antonio Plaza, *Fellow, IEEE*, Lizhe Wang, *Fellow, IEEE*, Zhonghui Tang, Li Zhuo, and Yuchen Yuan

***Abstract*— Supervised hyperspectral image classification suffers from the overfitting problem when limited labels are available. Graph-based semisupervised classifiers can tackle this problem by building connections between labeled and unlabeled samples. In this work, we prove the following two propositions for an optimal graph: 1) the interclass connection weights must be 0 and 2) for a given class, a subset must contain labeled samples or be connected to the remaining subset. In a semisupervised scenario, it is very difficult to ensure that the aforementioned propositions hold. Here, we introduce a new discriminant superpixel graph (dSPG) to build a suboptimal graph, which combines a newly proposed *within-superpixel* graph, aimed at disconnecting pixels belonging to different classes in a superpixel (so as to decrease interclass connection weights) and a *between-superpixel* graph that connects spectral adjacent superpixels (to increase the intraclass subset connections). We further propose a dSPG regularizer for hyperspectral image classification and a dSPG-guided graph convolutional network (dSPGCN) to extract discriminant features. Experimental results on real hyperspectral datasets demonstrate the good performance of our newly proposed dSPG for semisupervised hyperspectral image classification. The source codes for this study are available at https://github.com/yulong112/dSPG.**

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Long Yu is with the School of Geography and Planning, Sun Yat-sen University, Guangzhou 510275, China (e-mail: yulong110@gmail.com).

Jun Li and Lizhe Wang are with the School of Computer Science, China University of Geosciences, Wuhan 430078, China, and also with Hubei Key Laboratory of Intelligent Geo-Information Processing, China University of Geosciences, Wuhan 430078, China (e-mail: lijuncug@cug.edu.cn; Lizhe.Wang@gmail.com).

Lin He is with the School of Automation Science and Engineering, South China University of Technology, Guangzhou 510640, China (e-mail: helin@scut.edu.cn).

Antonio Plaza is with the Hyperspectral Computing Laboratory Department of Technology of Computers and Communications, Escuela Politécnica, University of Extremadura, 10003 Cáceres, Spain (e-mail: aplaza@unex.es).

Zhonghui Tang and Yuchen Yuan are with Zhongshan School of Medicine, Sun Yat-sen University, Guangzhou 510080, China (e-mail: tangzhh99@mail.sysu.edu.cn; yuanych9@mail2.sysu.edu.cn).

Li Zhuo is with Guangdong Provincial Key Laboratory of Urbanization and Geo-Simulation, Guangdong Provincial Engineering Research Center for Public Security and Disaster, School of Geography and Planning, Sun Yat-sen University, Guangzhou 510275, China, and also with the Southern Marine Science and Engineering Guangdong Laboratory, Zhuhai 519000, China (e-mail: zhuoli@mail.sysu.edu.cn).

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***Index Terms*— *Between-superpixel* graph, discriminant superpixel graph (dSPG), graph convolutional network (GCN), hyperspectral image classification, *within-superpixel* graph.**

I. Introduction

ONE of the main trends in hyperspectral image classification is supervised learning, in which supervised learners are trained by labeled training samples. Some of the most widely adopted traditional classifiers include the support vector machine (SVM) [1] and multinomial logistic regression (MLR) [2]. Besides, deep learning-based classification models [3], [4], [5], [6] have achieved superior performance due to their ability to extract nonlinear representations [7], [8]. Owing to the fact that the data labeling procedure is generally expensive and time-consuming [9], [10], semisupervised learning (exploiting both labeled and unlabeled samples to fit the classifiers) has become a well-known approach to improve the classification performance achieved by the labeled set [11].

The field of semisupervised learning for classification has developed rapidly [12], [13], [14], resulting in the emergence of many representative algorithms such as generative models [15], consistency regularization methods [16], [17], pseudo-labeling methods [18], and graph-based methods [7], [19], [20], [21]. To refine the classification, most of these algorithms exploit the associations between labeled and unlabeled samples and explore the real distribution of data. For example, the works [16], [17] argue that the model's predictions for both labeled and unlabeled samples obey a prior of smoothness. In particular, Miyato et al. [16] selectively smooth the output distribution of unlabeled samples in the adversarial direction. While in [18] and [22], unlabeled samples with high-quality pseudo labels are used to retrain or fine-tune the network. In graph-based methods, the associations between labeled and unlabeled samples can be directly modeled as the graph connections. The connection weight in a graph, reflecting the strength of the association between two samples, can be defined by explicit measures [21], or dynamically updated by graph learning methods [7]. Besides, other works in [23] combine self-supervision with semisupervised learning to obtain more robust features.

Among these algorithms, graph-based methods have recently received significant interest because graph structure

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naturally reflects the manifold assumption in semisupervised learning [14]. By utilizing relationships between samples, graph-based methods can capture structural information in the hyperspectral image. According to the graph structures, we group the available graph-based hyperspectral image classification methods into three major categories.

1) The first category considers pixel-level connections. In this category, the graph is built by connecting any two pixels in the hyperspectral image [18]. Examples include the nearest neighbor graph (NNG) [24] and others [25], [26]. NNG connects pixel pairs with high spectral similarities. These similarities are used as graph weights. Specifically, for a given pixel i, let $\mathbf{w}_{i^k} = [\mathrm{w}_{i,n_1}, \ldots, \mathrm{w}_{i,n_k}]$ be its graph weight vector holding the k largest weights, its connection with pixel j is defined as follows:

$$\tau_{i,j}^{\mathrm{NNG}} = \tau_{j,i}^{\mathrm{NNG}} = \begin{cases} 1, & \text{if } \mathrm{w}_{i,j} \in \mathbf{w}_{i^k} \text{ or } \mathrm{w}_{j,i} \in \mathbf{w}_{j^k} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

As it can be observed from (1), NNG relies only on spectral similarity to connect pixels and is thus susceptible to noise, leading to a large amount of interclass connections. To alleviate noise interference, NNGs based on band selection [27], sparse representation [25], and manifold representation [28] have been proposed successively. The methods based on manifold representation attempt to address the above problems by learning the nonlinear relationship of data. However, the sparse noise and spectral variations in the hyperspectral data still seriously affects the discriminability of spectral features, resulting in the interclass connections and bad classification results.

2) The second one improves pixel-level graphs with spatial information, leading to a spatial neighbor graph (SNG) [29], [30] and a spectral–spatial neighbor graph (SSNG) [31], [32], by using neighboring pixels. Let $\mathcal{N}(i)$ be the spatial neighborhood of pixel i, the connections of SNG and SSNG can be defined by the following equations:

$$\tau_{i,j}^{\mathrm{SNG}} = \begin{cases} 1, & \text{if } j \in \mathcal{N}(i) \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

$$\tau_{i,j}^{\mathrm{SSNG}} = \beta \tau_{i,j}^{\mathrm{NNG}} + (1-\beta)\tau_{i,j}^{\mathrm{SNG}} \tag{3}$$

where $0 \leq \beta \leq 1$ is a parameter to fuse the spectral and spatial neighborhoods. Although SNG smoothes the predictions of homogeneous image regions, it does not capture the long-range dependencies among pixels. Intuitively, this leads to many disconnected subsets in the same class, thus the global structure information of the hyperspectral image cannot be well preserved, and many regions may be misclassified. Moreover, both SNG and SSNG implicitly assume that spatially adjacent pixels may belong to the same class, even if in the boundary regions. Thus, similar to pixel-level graphs, SNG and SSNG also connect pixels belonging to different classes, resulting in interclass connection weights. Consequently, simply connecting spatially adjacent pixels is insufficient to achieve the desired outcome.

3) The third group intends to build superpixel-level graph (SPG) connections by taking each superpixel as a node [21], [33], [34], [35], [36]. In this group, some methods connect superpixel i and j only if they are spectral neighbors, that is, the superpixel version of (1) is adopted. For example, Hong et al. [37] propose a spatial–spectral manifold (SSM) regularization, constructing a graph matrix with both superpixels and pixels as nodes. SSM connects a superpixel node with its spectral adjacent superpixel nodes and with all pixel nodes within it. Although this method introduces local spatial information through superpixels, it is still easy to connect a large number of superpixels belonging to different classes. In addition, most other SPG-based methods connect spatially adjacent superpixels in the same way as SNG, or use (4) and its variants to combine superpixel-wise spatial and spectral neighbors

$$\tau_{i,j}^{\mathrm{SPG}} = \tau_{i,j}^{\mathrm{NNG}} \cdot \tau_{i,j}^{\mathrm{SNG}}. \tag{4}$$

Considering each superpixel as a node implies the assumption that all samples in a superpixel are connected and share the same class label. In real scenarios, a superpixel might contain multiple classes, especially in boundary regions. We name this phenomenon the mixing problem within superpixels. Therefore, the SPG contains interclass connections. On the other hand, the SPG connects spatially adjacent superpixels, which makes it difficult to ensure that intraclass subset connections are sufficient.

Motivated by these observations, in order to decrease the interclass connections and increase the intraclass subset connections, in this work we propose a new discriminant superpixel graph (dSPG) for hyperspectral image classification. First, we prove two propositions for an optimal graph.

1) The interclass connection weights must be 0.
2) For a given class, a subset must contain labeled samples or be connected to the remaining subset.

Due to the influence of noise, mixtures, and spectral variability, it is difficult to ensure the demands of the above two propositions. Thus, our proposed dSPG aims to build a suboptimal graph, introducing a *within-superpixel* graph and a *between-superpixel* graph to decrease the interclass connection weights and increase the intraclass subset connections. Specifically, we propose the following.

1) We define a *within-superpixel* graph that builds connections among the pixels within each superpixel, disconnecting pixels with high spectral differences inside a superpixel. Our aim is to decrease the interclass connections (as pixels with such high spectral differences likely belong to different classes).
2) We introduce a *between-superpixel* graph that is built by considering each superpixel as a node, connecting two superpixels if they are spectral adjacent (even if they are spatially disconnected), aiming at increasing the intraclass subset connections.

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With these two graphs, we achieve a discriminative suboptimal graph that approximates the objective of the defined optimal graph. We use dSPG for regularization in traditional semisupervised learning, and additionally develop a dSPG-guided graph convolutional network (dSPGCN) for deep feature extraction of hyperspectral cube. To explore the discriminant performance potential of dSPG sufficiently, we include a 1-D convolutional layer and a residual module in dSPGCN. The 1-D convolutional layer can mitigate the oversmoothing problem of graph convolutional network (GCN), while the residual module is used to improve the nonlinear representation capability of dSPGCN. Therefore, our proposed dSPGCN has the ability to extract more discriminant features. Furthermore, since we directly optimize the graph structure, the output graph can be applied not only for classification, but also for all graph-based applications such as hyperspectral image denoising.

The remainder of this article is organized as follows. Section II formulates the problem. Then, our newly proposed dSPG is presented in Section III. Section IV provides experimental results with a variety of hyperspectral images. Section V concludes the work and provides hints at plausible future research directions.

II. Problem Formulation

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n] \in \mathbb{R}^{b\times n}$ be the observed hyperspectral image data with $\mathbf{x}_i \in \mathbb{R}^b$ being the ith pixel, b being the number of spectral channels or bands, and n being the number of pixels. Let $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n]^\top \in \mathbb{R}^{n\times c}$ be the class label matrix with $\mathbf{y}_i = [y_i^{(1)}, \ldots, y_i^{(c)}]^\top \in \mathbb{R}^{c\times 1}$ being the label vector of the ith pixel and c being the number of classes. For a labeled pixel i with class label k, $y_i^{(k)} = 1$ and $y_i^{(j)} = 0$ for $j \neq k$. For any unlabeled pixel, $y_i^{(j)} = 0$ for $j = 1, \ldots, c$. With the aforementioned definitions in mind, we can build an undirected graph $G = \{\mathbf{X}, \mathbf{W}\}$ with $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_n] \in \mathbb{R}^{n\times n}$ being the graph symmetrical weight matrix of the connected edges between pixels, with $\mathbf{w}_i = [\mathrm{w}_{1,i}, \ldots, \mathrm{w}_{n,i}]^\top \in \mathbb{R}^{n\times 1}$ being the graph weight vector of the ith pixel. In general, $\mathbf{W}$ is built via specific spectral measures, such as the Gaussian kernel [33], cosine similarity [38], correlation, and covariance operators [39]. Therefore, larger values of $\mathrm{w}_{i,j}$ mean higher spectral similarity between pixels i and j and smaller values of $\mathrm{w}_{i,j}$ mean lower spectral similarity.

The traditional graph-based hyperspectral image classification, aimed at predicting class labels of unlabeled data based on the observation $\mathbf{X}$ and the labeled information in $\mathbf{Y}$, can be achieved as follows:

$$\begin{aligned}\widehat{\mathbf{F}} &= \arg\min_{\mathbf{F}} \|\mathbf{F}-\mathbf{Y}\|_F^2 + \mu\Omega(\mathbf{F}) \\ &= \arg\min_{\mathbf{F}} \|\mathbf{F}-\mathbf{Y}\|_F^2 + \mu\mathrm{tr}\left(\mathbf{F}^\top\mathbf{L}\mathbf{F}\right)\end{aligned} \tag{5}$$

where $\widehat{\mathbf{F}} \in \mathbb{R}^{n\times c}$ is the predicted label matrix, $\mu \geq 0$ is a regularization parameter, $\mathbf{I}$ is the identity matrix, and $\mathbf{L}$ is a Laplacian matrix with $\mathbf{L} = \mathbf{D} - \mathbf{W}$ in which $\mathbf{D}$ is a diagonal matrix with the ith diagonal element defined as $\mathbf{D}_{i,i} = \sum_j \mathrm{w}_{i,j}$. $\Omega(\mathbf{F}) = \mathrm{tr}(\mathbf{F}^\top\mathbf{L}\mathbf{F})$ is the graph regularizer in general graph-based methods, enforcing connected pixels to share the same class labels.

The optimization in (5) has a closed-form solution, which is as follows:

$$\widehat{\mathbf{F}} = (\mathbf{I} + \mu\mathbf{L})^{-1}\mathbf{Y}. \tag{6}$$

The final classification for pixel i is then computed as follows:

$$y_i^{(k)} = 1, \quad \text{with } k = \arg\max_k \widehat{\mathbf{F}}_{i,k}, \quad \text{for } k \in \{1, \ldots, c\}.$$

From (5) and (6), we can conclude that the performance of graph-based hyperspectral image classification is greatly affected by the graph regularizer $\Omega(\mathbf{F})$ defined by the graph structure $\mathbf{W}$.

III. Proposed Method

In this section, we first formulate the optimal graph for semisupervised classification. Then, we decompose the optimal graph into local and global parts, which are further realized by our proposed *within-superpixel* and *between-superpixel* graphs, respectively. Finally, based on these two graphs, we propose a dSPG regularizer and a dSPGCN for hyperspectral image classification.

A. Optimal Graph

Let $\mathbf{Y}_t$ be the truth label matrix. An optimal graph for classification is defined as the one producing the predicted label matrix as $\mathbf{F} = \mathbf{Y}_t$. Let the graph weights between pixels in the same class be intraclass weights, and those between pixels in different classes be interclass weights. In the following, we present two propositions for the evaluation of the optimal graph.

Proposition 1: For classification, the interclass weights of the optimal graph must be 0.

Proof: Let an optimal graph Laplacian matrix be defined as $\mathbf{L}_t$. Then, according to (6),

$$(\mathbf{I} + \mu\mathbf{L}_t)\mathbf{F} = \mathbf{Y}_t. \tag{7}$$

We divide all samples into two subsets: $\mathcal{S}_i$ (samples belonging to class i) and $\mathcal{S}_o$ (samples not belonging to class i). The optimal graph Laplacian matrix is represented as

$$\mathbf{L}_t = \begin{bmatrix} \mathbf{L}_t^i & \mathbf{L}_t^{i,o} \\ \mathbf{L}_t^{o,i} & \mathbf{L}_t^o \end{bmatrix}$$

where $\mathbf{L}_t^{i,o} = -\mathbf{W}^{i,o} = (\mathbf{L}_t^{o,i})^\top$ denote the graph weights between $\mathcal{S}_i$ and $\mathcal{S}_o$. Accordingly, $\mathbf{F}$ and the label matrix $\mathbf{Y}_t$ are represented as follows:

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}^i \\ \mathbf{F}^o \end{bmatrix}, \quad \mathbf{Y}_t = \begin{bmatrix} \mathbf{Y}_t^i \\ \mathbf{Y}_t^o \end{bmatrix}.$$

Let

$$\begin{aligned}\hat{\mathbf{L}}_t^i &= \mathbf{L}_t^i - \mu\mathbf{L}_t^{i,o}\left(\mathbf{I} + \mu\mathbf{L}_t^o\right)^{-1}\mathbf{L}_t^{o,i} \\ \hat{\mathbf{L}}_t^o &= \mathbf{L}_t^o - \mu\mathbf{L}_t^{o,i}\left(\mathbf{I} + \mu\mathbf{L}_t^i\right)^{-1}\mathbf{L}_t^{i,o}\end{aligned}$$

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the solution of (7) is

$$\mathbf{F}^i = \left(\mathbf{I} + \mu\hat{\mathbf{L}}_t^i\right)^{-1}\left[\mathbf{Y}_t^i - \mu\mathbf{L}_t^{i,o}\left(\mathbf{I} + \mu\mathbf{L}_t^o\right)^{-1}\mathbf{Y}_t^o\right] \tag{8a}$$

$$\mathbf{F}^o = \left(\mathbf{I} + \mu\hat{\mathbf{L}}_t^o\right)^{-1}\left[\mathbf{Y}_t^o - \mu\mathbf{L}_t^{o,i}\left(\mathbf{I} + \mu\mathbf{L}_t^i\right)^{-1}\mathbf{Y}_t^i\right]. \tag{8b}$$

As it can be observed from (8a), if $\mathbf{L}_t^{i,o} \neq \mathbf{0}$, the predicted results for samples in class i are influenced by those in the other classes, resulting in differences from the true labels. On the contrary, if $\mathbf{L}_t^{i,o} = \mathbf{0}$, the predicted labels for samples in class i are only determined by the label information of class i. Thus, for classification, the interclass weights (i.e., the weights in $\mathbf{W}^{i,o}$) of an optimal graph must be 0.

In real scenarios, the labels for classification are usually limited and it is not possible to ensure this condition. As it can be concluded from (8), an alternative option is to reduce $\mathbf{W}^{i,o}$, that is, decreasing the percentage of interclass weights $p^{i,o} = \sum \mathbf{W}^{i,o} / \sum \mathbf{W}$.

Proposition 2: Let the samples of a class be divided into any two subsets. For semisupervised classification, a subset in an optimal graph must contain labeled samples or be connected to the other subset.

Proof: According to Proposition 1, an optimal graph does not connect any samples belonging to different classes. Therefore, we only consider the one-class case. Let the two subsets be $s1$ and $s2$, respectively. There are two cases:

Case 1: both $s1$ and $s2$ have at least one labeled sample.

Case 2: $s1$ or $s2$ has no labeled samples.

The optimal graph Laplacian matrix is partitioned into

$$\mathbf{L}_t = \begin{bmatrix} \mathbf{L}_t^{s1} & \mathbf{L}_t^{s1,s2} \\ \mathbf{L}_t^{s2,s1} & \mathbf{L}_t^{s2} \end{bmatrix}$$

where $\mathbf{L}_t^{s1,s2} = -\mathbf{W}^{s1,s2} = (\mathbf{L}_t^{s2,s1})^\top$ denote the graph weights between $s1$ and $s2$. Accordingly, $\mathbf{F}$ and the initial label matrix $\mathbf{Y}$ are represented as follows:

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}^{s1} \\ \mathbf{F}^{s2} \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} \mathbf{Y}^{s1} \\ \mathbf{Y}^{s2} \end{bmatrix}.$$

Similarly, define the following two representations:

$$\hat{\mathbf{L}}_t^{s1} = \mathbf{L}_t^{s1} - \mu\mathbf{L}_t^{s1,s2}\left(\mathbf{I} + \mu\mathbf{L}_t^{s2}\right)^{-1}\mathbf{L}_t^{s2,s1}$$

$$\hat{\mathbf{L}}_t^{s2} = \mathbf{L}_t^{s2} - \mu\mathbf{L}_t^{s2,s1}\left(\mathbf{I} + \mu\mathbf{L}_t^{s1}\right)^{-1}\mathbf{L}_t^{s1,s2}$$

the solutions of (7) are

$$\mathbf{F}^{s1} = \left(\mathbf{I} + \mu\hat{\mathbf{L}}_t^{s1}\right)^{-1}\left[\mathbf{Y}^{s1} - \mu\mathbf{L}_t^{s1,s2}\left(\mathbf{I} + \mu\mathbf{L}_t^{s2}\right)^{-1}\mathbf{Y}^{s2}\right]$$

$$\mathbf{F}^{s2} = \left(\mathbf{I} + \mu\hat{\mathbf{L}}_t^{s2}\right)^{-1}\left[\mathbf{Y}^{s2} - \mu\mathbf{L}_t^{s2,s1}\left(\mathbf{I} + \mu\mathbf{L}_t^{s1}\right)^{-1}\mathbf{Y}^{s1}\right].$$

Case 1: If $\mathbf{Y}^{s1} \neq \mathbf{0}$ and $\mathbf{Y}^{s2} \neq \mathbf{0}$, as $\mathbf{L}_t^{s1,s2} \leq \mathbf{0}$, $\mathbf{F}^{s1} \neq \mathbf{0}$ and $\mathbf{F}^{s2} \neq \mathbf{0}$. Therefore, $\mathbf{F}$ is determined by the labels $\mathbf{Y}^{s1}$ and $\mathbf{Y}^{s2}$ in the same class.

Case 2: Let $s2$ be the subset with no labeled samples, i.e., $\mathbf{Y}^{s2} = \mathbf{0}$. The predicted results of $s1$ and $s2$ are represented as follows:

$$\mathbf{F}^{s1} = \left(\mathbf{I} + \mu\hat{\mathbf{L}}^{s1}\right)^{-1}\mathbf{Y}^{s1} \tag{9a}$$

$$\mathbf{F}^{s2} = -\mu\left(\mathbf{I} + \mu\hat{\mathbf{L}}^{s2}\right)^{-1}\mathbf{L}_t^{s2,s1}\left(\mathbf{I} + \mu\mathbf{L}_t^{s1}\right)^{-1}\mathbf{Y}^{s1}. \tag{9b}$$

As in (9b), if $\mathbf{L}_t^{s2,s1} = \mathbf{0}$, then $\mathbf{F}^{s2} = \mathbf{0}$. That is, if there is no connection between $s1$ and $s2$, $\mathbf{F}^{s2}$ cannot be predicted. To avoid this unpredictable case, any intraclass subset in an optimal graph must contain labeled samples or be connected to the remaining subset.

We can conclude that the classification performance improves when decreasing the percentage of interclass weights and increasing the connections between intraclass subsets.

B. Superpixel Assumptions

In a hyperspectral image without noise, mixtures, and spectral variability, there should be no variability within the same class and no cross similarity between samples of different classes. If we perform superpixel segmentation for the hyperspectral image, we can assume that each superpixel has at least one label. We can build an optimal graph by considering each superpixel as a node for the following reasons.

1) The connections among samples within a superpixel perfectly adjust to Proposition 1.
2) The connections between superpixels belonging to the same class perfectly adjust to Case 1 of Proposition 2.

However, in real scenarios, spectral variations cannot be ignored. The situation gets worse with noise and mixtures [40]. All this leads to the fact that more than one class is possible in a superpixel. Furthermore, it is also difficult to meet the requirement of having at least one label in a superpixel. When there are no labels, it is difficult to ensure the connection demanded by Case 2 of Proposition 2.

Therefore, in this work, we build a suboptimal graph by decreasing the interclass weights and increasing the intraclass subset connections, achieved by two newly proposed concepts: the *within-superpixel* graph and the *between-superpixel* graph.

1) The *within-superpixel* graph connects the samples within each superpixel. The connections are mostly between samples of the same class, since (based on spatial consistency) most samples within the superpixel belong to the same class. Thus, in comparison with pixel-level-based graphs, the *within-superpixel* graph decreases the interclass weights, in response to Proposition 1.
2) The *between-superpixel* graph builds connections between superpixels by considering each superpixel as a node, therefore increasing the connections between intraclass subsets, in response to Proposition 2.

C. Within-Superpixel Graph

The *within-superpixel* graph defines a sparse graph inside superpixels. First, for any two pixels $\mathbf{x}_i = [x_{i,1}, \ldots, x_{i,b}]^\top$ and $\mathbf{x}_j = [x_{j,1}, \ldots, x_{j,b}]^\top$, we define

$$\mathbf{w}_{i,j} = \left| \frac{\sum_{k=1}^b \left(x_{i,k} - \overline{x}_i\right)\left(x_{j,k} - \overline{x}_j\right)}{\sqrt{\sum_{k=1}^b \left(x_{i,k} - \overline{x}_i\right)^2}\sqrt{\sum_{k=1}^b \left(x_{j,k} - \overline{x}_j\right)^2}} \right| \tag{10}$$

where $\overline{x}_i = \sum_{k=1}^b x_{i,k}$ and $\overline{x}_j = \sum_{k=1}^b x_{j,k}$.

Then, assume that $\mathcal{T} = [\boldsymbol{\tau}_1, \ldots, \boldsymbol{\tau}_n] \in \mathbb{R}^{n\times n}$ is a sparse connection matrix, with $\boldsymbol{\tau}_i = [\tau_{i,1}, \ldots, \tau_{i,n}] \in \mathbb{R}^n$ being the sparse connection vector for pixel i. Notice that $\tau_{i,j} = 1$ for

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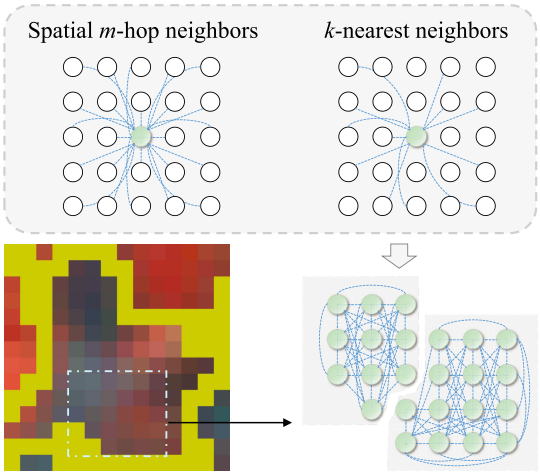

Fig. 1. Diagram of *within-superpixel* graph. Each pixel is connected to its superpixel-bounded spatial m-hop neighbors and k-nearest neighbors.

$j = 1, \ldots, n$, which indicates that there is a connection between pixels i and j, and $\tau_{i,j} = 0$ otherwise. The sparse graph weight matrix $\overline{\mathbf{W}}$ can be obtained as follows:

$$\overline{\mathbf{W}} = \mathbf{W} \odot \mathcal{T}$$

where $\odot$ stands for the Hadamard product of matrices.

We first introduce the *within-superpixel* graph exploiting the connections between pixels inside superpixels. For pixels within the same superpixel, we build the *within-superpixel* graph, $\overline{\mathbf{W}}^{\text{ws}}$, as follows:

$$\overline{\mathbf{W}}^{\text{ws}} = \mathbf{W} \odot \mathcal{T}^{\text{ws}} \tag{11}$$

where $\mathcal{T}^{\text{ws}}$ denotes the sparse connections between pixels within the superpixel. For any pixel $\mathbf{x}_i$, assuming that $\mathbf{x}_i \in \text{sp}(s)$, let $\mathbf{w}_{i^k}$ be its k-nearest neighbor weights, and let $\mathcal{N}^s(i)$ be its superpixel-bounded spatial neighborhood. That is, only the spatial neighboring pixels of $\mathbf{x}_i$ located in $\text{sp}(s)$ are considered for $\mathcal{N}^s(i)$. Then, we have

$$\tau_{i,j}^{\text{ws}} = \begin{cases} 1, & \text{if } \text{w}_{i,j} \in \mathbf{w}_{i^k},\ j \in \mathcal{N}^s(i) \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where $\mathcal{N}^s(i) = \{\mathbf{x}_u : ||\text{row}(u) - \text{row}(i)||_2 \leq m \cap ||\text{col}(u) - \text{col}(i)||_2 \leq m \cap \mathbf{x}_u \in \text{sp}(s)\}$, $\text{row}(i)$ and $\text{col}(i)$ represent the row index and column index of pixel $\mathbf{x}_i$ in the hyperspectral image, respectively, and m indicates that the spatial neighbors are within the m-hop. As shown in (12) and Fig. 1, inside a superpixel, pixels with high spectral differences are likely disconnected. Considering that pixels with high spectral differences likely belong to different classes, the proposed disconnection inside superpixels therefore decreases the interclass weights. Therefore, the *within-superpixel* graph can deal with the mixing problem that there may be multiple classes of samples within a superpixel.

D. Between-Superpixel Graph

The *between-superpixel* graph defines a sparse graph between superpixels. Different from the existing superpixel graphs [33], [34], we regard a superpixel as a predefined cluster of samples. To avoid a huge increase in the interclass connection weights, we only connect the cluster centers of superpixels. The determination of cluster centers is usually an important step [41]. Here, we select the most representative pixel in a superpixel as its cluster center. Specifically, we introduce a new concept: *the most representative weight* of two superpixels. Let $w(t, s)$ be the weight between the median feature of $\text{sp}(t)$ and that of $\text{sp}(s)$. We define $w(t, s)$ to be the most representative weight. Notice that, with the median feature, the most representative weights between superpixels are robust against both Gaussian noise and salt-and-pepper noise. Accordingly, we define $\mathbf{x}(i)$ and $\mathbf{x}(j)$, with

$$\mathbf{x}(i), \mathbf{x}(j) = \underset{\mathbf{x}(i) \in sp(t), \mathbf{x}(j) \in sp(s)}{\arg\min} \left| \mathbf{w}_{i,j} - w(t, s) \right|$$

being the most representative pixels of $\text{sp}(t)$ and $\text{sp}(s)$, respectively.

Let $\overline{\mathbf{W}}^{\text{bs}}$ denote the sparse connections between superpixels. Specifically, we define

$$\overline{\mathbf{W}}^{\text{bs}} = \mathbf{W} \odot \mathcal{T}^{\text{bs}} \tag{13}$$

where $\mathcal{T}^{\text{bs}}$ denotes the sparse connections between superpixels.

In the following, we present the adopted strategy to obtain $\mathcal{T}^{\text{bs}}$. Recall that $\mathbf{W}^c \in \mathbb{R}^{\text{sp}_n \times \text{sp}_n}$ is a contracted weight matrix for superpixels with

$$w_{s,t}^c = w(s, t).$$

Let $\mathcal{T}^c$ be the corresponding contracted sparse connection between neighboring superpixels. Let $\mathbf{w}_{s^k}^c = [\text{w}_{s,n_1}^c, \ldots, \text{w}_{s,n_k}^c]$ be the vector holding the k largest weights. Here, $\text{sp}(t)$ is considered to be a neighbor of $\text{sp}(s)$ if $\text{w}(s, t) \in \mathbf{w}_{s^k}^c$. Then, we can obtain

$$\tau_{s,t}^c = \begin{cases} 1, & \text{if } \text{w}(s, t) \in \mathbf{w}_{s^k}^c \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

Notice that $\mathcal{T}^c$ may be asymmetric. For any two superpixels s and t, $\text{w}(s, t)$ may not be in both $\mathbf{w}_{s^k}^c$ and $\mathbf{w}_{t^k}^c$. Therefore, in order to ensure that $\mathcal{T}^c$ is symmetric, we perform

$$\mathcal{T}^c = \mathcal{T}^c \odot \left(\mathcal{T}^c\right)^\top \tag{15}$$

which produces mutual connections among superpixels with the advantage that the solution exhibits high sparsity and mutual agreement [42]. However, it might lead to a situation that there are some isolated superpixels, i.e., superpixels without connections to any other ones. In this case, we enforce $\tau_{s,k_1}^c = \tau_{k_1,s}^c = 1$.

Then, we obtain the superpixel sparse connection matrix. Let $\mathcal{N}(\mathbf{x}_i) = \{\mathbf{x}_u : ||\text{row}(u) - \text{row}(i)||_2 \leq m \cap ||\text{col}(u) - \text{col}(i)||_2 \leq m \cap \mathbf{w}_{i,u} > \varepsilon\}$ be the neighborhood of pixel $\mathbf{x}_i$, where ε is a constant that is empirically set to filter higher weights. For any two pixels $\mathbf{x}_i$ and $\mathbf{x}_j$, the following conditions are considered:

1) If they belong to the same superpixel,

$$\tau_{i,j}^{\text{bs}} = 0.$$

2) If they belong to two different superpixels, say, $\mathbf{x}_i \in \text{sp}(s)$ and $\mathbf{x}_j \in \text{sp}(t)$, respectively, when $\tau_{s,t}^c = 0$,

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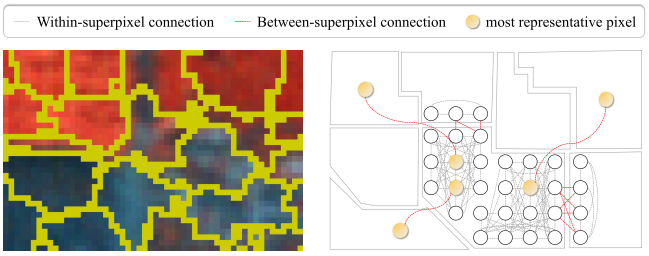

Fig. 2. Diagram of dSPG, in which *between-superpixel* graph contains the connections between spatial adjacent superpixels and the long-range connections.

we analyze the connection from the following two viewpoints.

a) In the first case, when $\mathbf{x}_j \in \mathcal{N}(\mathbf{x}_i)$, we have

$$\tau_{i,j}^{\text{bs}} = 1.$$

b) Otherwise, we set

$$\tau_{i,j}^{\text{bs}} = 0.$$

3) Finally, if $\mathbf{x}_i \in \text{sp}(s)$ and $\mathbf{x}_j \in \text{sp}(t)$, and $\tau_{s,t}^{c} = 1$, we analyze the connection from the following three viewpoints.

a) In the first case, when $\mathbf{x}_i = \mathbf{x}(s)$ and $\mathbf{x}_j = \mathbf{x}(t)$, that is, $\mathbf{x}_i$ and $\mathbf{x}_j$ are the most representative pixels of their respective superpixels, we define

$$\tau_{i,j}^{\text{bs}} = 1.$$

b) In the second case, when $\mathbf{x}_j \in \mathcal{N}(\mathbf{x}_i)$, we have

$$\tau_{i,j}^{\text{bs}} = 1.$$

c) Otherwise, we set

$$\tau_{i,j}^{\text{bs}} = 0.$$

In summary, we define the *between-superpixel* connections for a given pair of pixels as follows.

1) If $\mathbf{x}_i \in \text{sp}(s)$, $\mathbf{x}_j \in \text{sp}(t)$,

$$\tau_{i,j}^{\text{bs}} = \begin{cases} 1, & \text{if } \tau_{s,t}^{c} = 1, \mathbf{x}_i = \mathbf{x}(s), \mathbf{x}_j = \mathbf{x}(t) \quad (16a) \\ 1, & \text{if } \mathbf{x}_j \in \mathcal{N}(\mathbf{x}_i). \quad (16b) \end{cases}$$

2) Otherwise

$$\tau_{i,j}^{\text{bs}} = 0. \quad (17)$$

As it can be observed, the proposed *between-superpixel* graph connects superpixels from the viewpoint of spectral information [see (16a)]. By connecting the most representative pixels using (16a), the proposed *between-superpixel* graph connects two superpixels if they are spectral adjacent. This is particularly important for situations in which some of the isolated regions do not have labeled samples, as it increases the connections between intraclass subsets and captures the long-range dependencies among superpixels.

E. Discriminant Superpixel Graph Regularizer

With the proposed *within-superpixel* graph and *between-superpixel* graphs, we build a suboptimal graph $\overline{\mathbf{W}}$—namely, dSPG—as follows:

$$\overline{\mathbf{W}} = (1-\alpha)\overline{\mathbf{W}}^{\text{ws}} + \alpha\overline{\mathbf{W}}^{\text{bs}} \quad (18)$$

where α is the parameter controlling the degree between $\overline{\mathbf{W}}^{\text{ws}}$ and $\overline{\mathbf{W}}^{\text{bs}}$. As illustrated in Fig. 2, dSPG can assign multiple class labels to a mixed superpixel by connecting different points in a superpixel and disconnecting pixels with large spectral differences within the superpixel.

Let $\overline{\mathbf{L}} = \overline{\mathbf{D}} - \overline{\mathbf{W}}$ denote the Laplacian matrix of our proposed suboptimal graph, where $\overline{\mathbf{D}}$ is the degree matrix. The graph regularizer in (5) turns to

$$\begin{aligned} \Omega(\mathbf{F}) &= \text{tr}\{\mathbf{F}^{\top}\overline{\mathbf{L}}\mathbf{F}\} \\ &= \frac{1}{2}\sum_{k=1}^{c}\sum_{i,j=1}^{n}\overline{w}_{i,j}\left(\mathbf{F}_{i,k} - \mathbf{F}_{j,k}\right)^2. \end{aligned} \quad (19)$$

Combing (18) and (19), we have

$$\Omega_d(\mathbf{F}) = \frac{1}{2}\left\{\begin{array}{l} (1-\alpha)\displaystyle\sum_{k=1}^{c}\sum_{i,j=1}^{n}\overline{w}_{i,j}^{\text{ws}}\left(\mathbf{F}_{i,k} - \mathbf{F}_{j,k}\right)^2 \\ +\alpha\displaystyle\sum_{k=1}^{c}\sum_{i,j=1}^{n}\overline{w}_{i,j}^{\text{bs}}\left(\mathbf{F}_{i,k} - \mathbf{F}_{j,k}\right)^2 \end{array}\right\} \quad (20)$$

where $\overline{w}_{i,j}^{\text{ws}}$ and $\overline{w}_{i,j}^{\text{bs}}$ are the weights in $\overline{\mathbf{W}}^{\text{bs}}$ and $\overline{\mathbf{W}}^{\text{ws}}$, respectively.

F. dSPGCN

Before training GCN, to capture global higher order neighbors, refer to [43], we perform a graph learning process on dSPG. Mathematically, this process can be interpreted as follows:

$$\min_{\mathbf{A}} \frac{\mu_1}{2}\sum_{r=1}^{n}\sum_{i,j=1}^{n}\overline{\mathbf{W}}_{ij}\left\|\frac{\mathbf{A}_{ri}}{\sqrt{\overline{\mathbf{D}}_{ii}}} - \frac{\mathbf{A}_{rj}}{\sqrt{\overline{\mathbf{D}}_{jj}}}\right\|^2 + \|\mathbf{A}-\mathbf{I}\|^2 \quad (21)$$

where $\overline{\mathbf{W}}_{ij}$ denotes similarity between pixels i and j, $\mathbf{I}$ is the identity matrix, and $\mathbf{A} \in \mathbb{R}^{n\times n}$ is the desired weight matrix. In (21), the first term requires that neighboring vertices in the graph share similar topological relationships. The second term enforces a penalty on the connection weights between different vertices, thereby limiting the density of graph connections. These two terms are traded off by the parameter μ_1. The closed-form solution of (21) is given as follows:

$$\mathbf{A} = \frac{1}{\mu_1+1}\left(\mathbf{I} - \frac{\mu_1}{\mu_1+1}\overline{\mathbf{D}}^{-\frac{1}{2}}\overline{\mathbf{W}}\overline{\mathbf{D}}^{-\frac{1}{2}}\right)^{-1}. \quad (22)$$

$\mathbf{A}$ not only inherits the discriminative ability of dSPG, but also captures the high-order adjacency relationship between pixels. With $\mathbf{A}$, we construct the dSPGCN, as illustrated in Fig. 3. First, we input all data $\mathbf{X}^{\top} \in \mathbb{R}^{n\times b}$ into a 1-D convolutional layer with a kernel size of 10, generating an output $\mathbf{H}^{(0)} \in \mathbb{R}^{n\times 4b}$. The purpose of performing the 1-D convolutional layer is to improve the discriminability of

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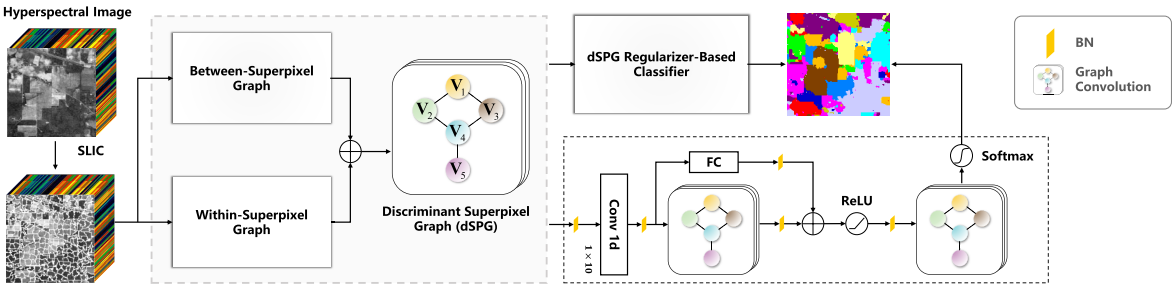

Fig. 3. Overall architecture of our proposed dSPG-based hyperspectral image classification. First, the dSPG is generated by combining a *within-superpixel* graph and a *between-superpixel* graph. Second, we designed the dSPG regularizer-based classifier and a dSPG-guided GCN (dSPGCN). In dSPGCN, we proposed an innovative combination of 1-D convolutional layer, residual module, and graph convolutional layers to excavate the potential of the dSPG for extracting the discriminant features underlying in the hyperspectral image.

features and alleviate the oversmoothing problem of GCN. Then, we feed $\mathbf{H}^{(0)}$ into the network consisting of two graph convolutional layers and a residual module. The dSPGCN model can be mathematically defined as (23)

$$\begin{cases} \mathbf{H}^{(0)} = \mathrm{BN}(\mathrm{Conv1d}(\mathrm{BN}(\mathbf{X}^{\top}))) & \text{(23a)} \\ \mathbf{H}^{(1)} = \mathrm{BN}\Big(\mathrm{ReLU}\Big(\tilde{\mathbf{S}}\mathbf{H}^{(0)}\boldsymbol{\Theta}_0^{(0)} + \mathbf{H}^{(0)}\boldsymbol{\Theta}_1^{(0)}\Big)\Big) & \text{(23b)} \\ \mathbf{H}^{(2)} = \mathrm{soft\,max}\big(\tilde{\mathbf{S}}\mathbf{H}^{(1)}\boldsymbol{\Theta}^{(1)}\big) & \text{(23c)} \end{cases}$$

where $\tilde{\mathbf{S}} = \tilde{\mathbf{D}}^{-1/2}(\mathbf{A}+\mathbf{I})\tilde{\mathbf{D}}^{-1/2}$ denotes the renormalized variant of $\mathbf{A}$ and $\tilde{\mathbf{D}}_{ii} = \sum_{j=1}^{n}(\mathbf{A}+\mathbf{I})_{ij}$. $\boldsymbol{\Theta}_0^{(0)}$ and $\boldsymbol{\Theta}_1^{(0)}$ are trainable parameters of the first graph convolutional layer and the fully connected (FC) layer in the residual module, respectively, mapping the $4b$-dimensional features of $\mathbf{H}^{(0)}$ to b_1-dimensional features. $\boldsymbol{\Theta}^{(1)}$ is the parameter of the second graph convolutional layer. BN$(\cdot)$ denotes the batch normalization layer. The final output of dSPGCN is defined as $\mathbf{Z} = \mathbf{H}^{(2)}$ and the classification loss is measured by cross entropy.

As two classification frameworks, dSPG and dSPGCN each have their own advantages, which are listed as follows.

1) *Efficiency:* The dSPG regularizer-based classifier is better because it performs mathematical calculations directly without training, allowing for fast predictions.
2) *Extensibility:* The dSPGCN-based deep learning framework is superior because it can learn discriminant features and apply them to other tasks, which dSPG regularizer-based classifier cannot.
3) *Robustness:* The dSPG regularizer-based classifier is more robust to noise, while dSPGCN is more susceptible to data quality.
4) *Application Scenarios:* The dSPG regularizer-based classifier is more suitable for cases with few labeled samples, as it relies only on the local information in the graph structure and does not need to learn from high-dimensional features. When there are sufficient labels, dSPGCN is likely to achieve better ground target classification than dSPG regularizer-based classifier.

IV. Experimental Results

In this section, we evaluate the effectiveness of our proposed dSPG method using four hyperspectral images (collected by different earth observation instruments such as airplanes, satellites, and unmanned aerial vehicles).

1) The Indian Pines image [44] was gathered by the airborne visible infrared imaging spectrometer (AVIRIS) [45] over a mixed agriculture–forest scenario. In the considered scene, see Fig. 4(a), there are 145 × 145 pixels, 200 spectral bands, and 12 classes, with a spatial resolution of 20 m and a total of 10 171 samples, as shown in Table I. We retain samples of these 12 classes in order to avoid small sample size problems. In this experimental setting, we can fairly compare the classification performance of different graph structures under a large number of labeled samples. The ground-truth image is given in Fig. 4(b). As it can be observed from Fig. 4, the classes are distributed into small regions with high discontinuity. This means that there are many superpixels located at/near class boundaries. Therefore, the AVIRIS dataset can be used to validate the effectiveness of the proposed approach in scenarios with high discontinuity.
2) The Zaoyuan dataset, see Fig. 5(a), was collected by the operational modular imaging spectrometer (OMIS) and contains 80 spectral bands with a spatial resolution of 3.6 m. The Zaoyuan scene consists of 137 × 202 pixels (23 821 of which have reference labels) and can be divided into eight classes. These ground-truth classes are displayed in Fig. 5(b) and Table I. The OMIS dataset exhibits class homogeneity, in the sense that each class is located in one single region. This means that neighboring superpixels are more likely belonging to the same class. On the other hand, as each homogenous region is likely to be segmented into many superpixels, there are many distant superpixels belonging to the same class. Therefore, the OMIS dataset aims at validating the effectiveness of the proposed approach in scenarios with high spatial homogeneity.
3) The HongHu dataset was acquired by the hyperspectral imager on the ZiYuan-1 (02D) [ZY-1(02D)] satellite [46], over a rural–urban fringe area in Honghu, and contains 166 spectral bands from 0.4 to 2.5 μm and 119 × 374 pixels with a spatial resolution of 30 m. Before experiments, we removed the water absorption and low signal-to-noise ratio bands, leaving 127 bands for subsequent classification. We labeled 6807 pixels in this dataset into five classes (Table X): Vegetation, Pond, Impervious surface, River, and Cropland.

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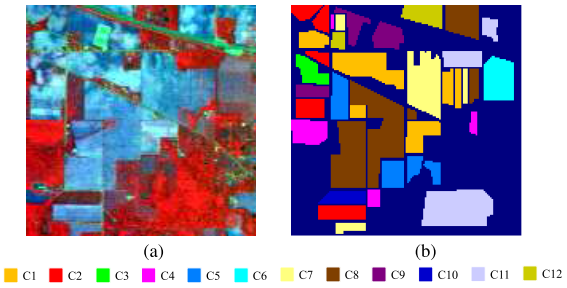

Fig. 4. AVIRIS Indian Pines dataset. (a) False color composition. (b) Ground truth.

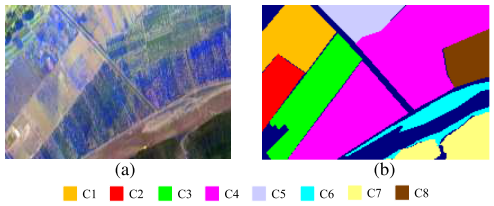

Fig. 5. OMIS Zaoyuan dataset. (a) False color composition. (b) Ground truth.

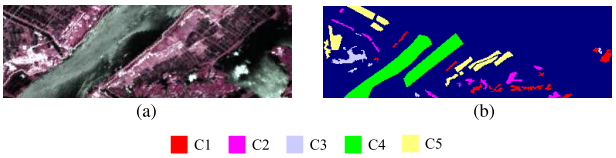

Fig. 6. ZY-1(02D)-HongHu satellite dataset. (a) False color composition. (b) Ground truth.

We labeled 5426 pixels in this dataset into five classes: *Vegetation*, *Pond*, *Impervious surface*, *River*, and *Cropland*. These pixels are used as reference pixels for selecting the labeled training set and calculating classification accuracy. As it can be seen from the false color image in Fig. 6, this scene is characterized by a large number of spatially fragmented land cover regions. The challenge is to capture the exact boundaries of the land cover areas.

4) The WHU-Hi dataset [47], [48] was gathered over a complex agricultural zone by the Headwall Nano-Hyperspec sensor on an unmanned aerial vehicle (drone). As shown in Fig. 7, this dataset contains 114 $\times$ 337 pixels with a spatial resolution of 0.043 m. Each pixel has 270 bands from 0.4 to 1 μm. In this dataset, 34 169 pixels are labeled into 12 classes for reference (Table X). The challenge of this dataset is to achieve fine-grained classification of crops.

In our experiments, we use the overall accuracy (OA) to evaluate the classification accuracy. In order to guarantee statistical significance, each reported OA value is obtained as the average of 10 Monte Carlo runs.

A. Parameter Analysis

There are four parameters involved in the proposed method, which are set as follows.

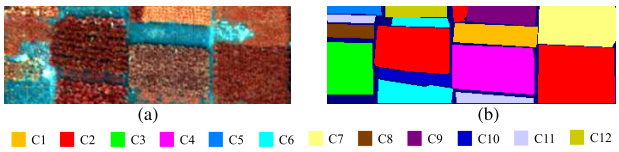

Fig. 7. WHU-Hi unmanned aerial vehicle (drone) dataset. (a) False color composition. (b) Ground truth.

TABLE I

GROUND TRUTH AND NUMBER OF LABELED SAMPLES IN EACH CLASS FOR THE AVIRIS AND OMIS HYPERSPECTRAL IMAGES

AVIRIS			OMIS		
Class	Name	Pixels	Class	Name	Pixels
C1	Corn-notill	1434	C1	Vegetable	2625
C2	Corn-mintill	834	C2	Grape	1302
C3	Corn	234	C3	Dry vegetable	3442
C4	Grass-pasture	497	C4	Pear	10243
C5	Grass-trees	747	C5	Corn	1425
C6	Hay-windrowed	489	C6	Terrace/Grass	1484
C7	Soybean-nottill	968	C7	Bush-Lespedeza	1808
C8	Soybean-mintill	2468	C8	Peach	1492
C9	Soybean-clean	614			
C10	Wheat	212			
C11	Woods	1294			
C12	Buildings-Grass-Trees-Drives	380			

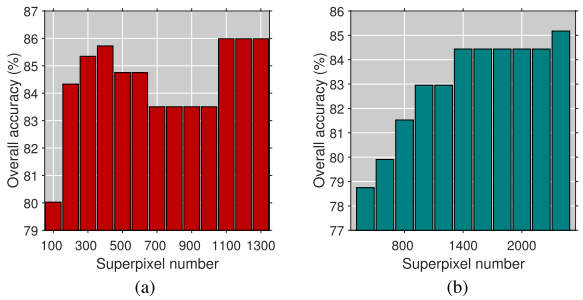

Fig. 8. Sensitivity analysis of parameter sp_n in our proposed method using the AVIRIS and OMIS datasets. (a) AVIRIS. (b) OMIS.

1) The regularization parameter μ in the graph-based classifier in (5), which is selected by empirical testing in a coarse-to-fine search method, following [33].
2) The parameter k considered to define the spectral nearest neighborhood, which is set to 10 empirically.
3) The number of superpixels sp_n.
4) The graph fusion parameter α in (18).

As mentioned before, the first two parameters are set empirically. In fact, we found out that our proposed approach is insensitive to these two parameters. Therefore, we focus on how to set the other two parameters.

1) Impact of the Number of Superpixels: We first analyze the effect of the number of superpixels sp_n on the obtained classification results. Fig. 8 shows the obtained OAs for the AVIRIS and OMIS datasets by using five labeled samples per class.

1) From Fig. 8(a) we can observe that, for the AVIRIS dataset, when $sp_n \geq 1100$ the obtained OAs are remarkable and stable as the number of superpixels increases. Therefore, in subsequent experiments we choose $sp_n = 1100$ for the AVIRIS dataset, for the sake of computational cost.
2) For the OMIS dataset, when $sp_n \geq 1400$ the proposed approach achieves very good results. Therefore,

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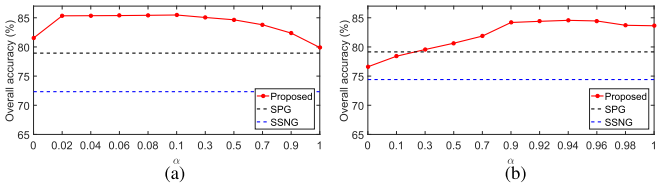

Fig. 9. Sensitivity analysis of parameter α in our proposed method using the AVIRIS and OMIS datasets. (a) AVIRIS. (b) OMIS.

in subsequent experiments we choose $\text{sp}_n = 1400$ for the OMIS dataset.

2) Impact of the Fusion Parameter α: Fig. 9 plots the OA achieved by our dSPG as a function of α by using five labeled samples per class. For comparison, the results obtained by SPG and SSNG are also reported. From Fig. 9, we can observe that the proposed method achieved the best results.

1) For the AVIRIS dataset, it can be observed from Fig. 9(a) that the proposed approach obtains the best results when $0 < \alpha \leq 0.3$. This indicates that the *within-superpixel* graph played an important role in this scene. This is expected, since this dataset has many boundaries, with the disconnections by the proposed *within-superpixel* graph having a high impact on the results. In subsequent experiments, we set $\alpha = 0.1$ for the AVIRIS dataset.
2) For the OMIS dataset, it can be observed from Fig. 9(b) that the proposed approach obtains very good results when $0.8 \leq \alpha < 1$. It is clear that, for this dataset, the *between-superpixel* graph plays a more important role. This is also consistent with our observations on this dataset [Fig. 5(c)]: as there is high spatial homogeneity in the available classes, it is more important to increase intraclass subset connections. In subsequent experiments, we set $\alpha = 0.9$ for the OMIS dataset.

B. Quantitative Comparison With State-of-the-Art Approaches

We perform a comparison of our dSPG against some state-of-the-art methods, including traditional graph-based methods (NNG, SNG, SSNG, SSM [37], and SPG [33]), GCN-based methods (MDGCN [49], DMSGer [7], GiGCN [21], MSSGU [36], and CEGCN [35]), nongraph-based deep learning methods (3-D CNN [50], HybridSN [51], Gabor-Nets [3], PCNN [52], 3DOC-SSAN [53], LWCNN [54], APDCLNN [55], and SpectralFormer [56]), and traditional supervised classifiers (SVM [1] and MLR [2]). When comparing with the deep learning methods on the AVIRIS dataset, we used all 16 classes to ensure applicability to real-world scenarios. Moreover, inspired by GiGCN [21], we replace the *Internal-Graph* in GiGCN with our proposed *within-superpixel* graph and the *External-Graph* with our proposed *between-superpixel* graph to form a dSPG-based graph-in-GCN (dSPGiGCN) for additional comparison. We use very small sets of randomly selected labeled pixels (3, 5, 7, 10, 15, 20, and 50 per class) for evaluation. The obtained OAs are given in Tables II–VI for the AVIRIS and OMIS datasets, respectively.

Without loss of fairness, we first verify the hyperspectral image classification performance of dSPG and other graph structures (NNG, SNG, SSNG, SSM [37], and SPG [33]) under the framework of (5). As it can be observed from Table II, the proposed method achieves the best classification accuracies on the AVIRIS dataset in all cases. The advantages are more significant when the sets of labeled samples are small. For instance, when the number of labeled pixels is only 3 per class, the OA obtained by the proposed approach is around 6% higher than that obtained by SPG, which is the second best, and more than 11% better than those obtained by the remaining competitors. In order to achieve an OA of around 85%, the proposed approach only needs five labeled samples per class, while the SPG needs at least ten, the SNG needs at least 15, and the remaining methods need more than 20. This is a significant advantage, revealing that the proposed approach can better tackle classification problems with limited labeled samples. From Table III we can conclude that, compared with nongraph-based deep learning methods, the classification accuracies of our proposed dSPG and dSPGCN are greatly improved under a small number of labeled samples. Compared with other GCN methods (MDGCN, DMSGer, and GiGCN), the proposed dSPG has a significant advantage under less than 50 labeled samples per class. As it can be seen from Table IV, with ten labeled samples in each class, the Kappa of dSPG is 5.6% higher than that of GiGCN. When there are

TABLE II

OVERALL ACCURACY [%] OBTAINED BY DIFFERENT METHODS FOR THE AVIRIS DATASET (STANDARD DEVIATIONS IN THE PARENTHESES)

Number of labeled pixels per class	**dSPG**	SPG	SSM	SSNG	SNG	NNG	SVM	MLR
3	**78.08**	71.67	54.60	66.61	63.36	51.13	52.65	53.98
	(5.51)	(2.99)	(6.92)	(4.02)	(3.00)	(5.07)	(5.29)	(4.07)
5	**85.48**	78.94	64.96	72.33	72.80	56.33	60.72	59.91
	(3.62)	(2.40)	(5.47)	(3.35)	(2.81)	(4.86)	(3.32)	(3.88)
7	**88.11**	82.37	69.47	75.78	77.29	59.64	66.09	64.08
	(2.70)	(2.08)	(3.47)	(3.84)	(2.52)	(3.73)	(1.18)	(3.20)
10	**89.72**	84.64	73.93	78.34	81.24	62.40	68.45	66.85
	(1.95)	(2.02)	(3.18)	(3.92)	(2.02)	(3.21)	(1.76)	(2.75)
15	**91.21**	87.40	78.56	80.40	85.15	64.63	70.91	69.16
	(1.59)	(2.21)	(2.20)	(2.51)	(2.36)	(2.53)	(1.60)	(1.86)
20	**92.91**	89.27	80.92	81.76	87.88	65.96	73.29	71.45
	(1.65)	(2.52)	(1.88)	(1.99)	(1.94)	(2.37)	(1.26)	(1.26)
50	**96.22**	94.44	88.55	86.60	94.29	70.07	80.21	78.75
	(0.55)	(1.12)	(1.57)	(1.03)	(0.83)	(1.34)	(1.17)	(1.00)

TABLE III

ACCURACIES [%] OBTAINED WITH DIFFERENT LABELED SAMPLES PER CLASS, RESPECTIVELY, RANDOMLY SELECTED FROM THE GROUND TRUTH OF 16 CLASSES FOR THE AVIRIS DATASET. * MARKS THE METHODS IMPLEMENTED BY OTHER AUTHORS, AND - MARKS NO REPORTED RESULT

	10/class	30/class	50/class	100/class
3-D CNN* [50]	-	-	88.78 ± 0.78	95.05 ± 0.28
HybridSN [51]	76.75 ± 1.37	90.60 ± 1.53	93.96 ± 0.87	97.47 ± 0.38
Gabor-Nets* [3]	-	-	94.05 ± 0.79	97.01 ± 0.52
PCNN* [52]	74.47	87.30	-	-
3DOC-SSAN [53]	64.06 ± 7.69	87.02 ± 2.08	91.83 ± 1.44	96.19 ± 0.81
LWCNN [54]	74.02 ± 2.16	86.74 ± 1.86	91.55 ± 1.06	95.24 ± 1.02
APDCLNN* [55]	77.82	92.36	-	-
SpectralFormer [56]	48.01 ± 2.03	72.80 ± 2.76	80.15 ± 1.80	89.46 ± 1.75
MDGCN [49]	75.25 ± 3.01	87.50 ± 0.91	90.62 ± 1.41	93.99 ± 0.72
DMSGer [7]	78.77 ± 1.37	91.33 ± 1.14	94.41 ± 0.79	97.26 ± 0.49
GiGCN [21]	84.66 ± 2.30	92.72 ± 0.80	94.58 ± 0.68	95.73 ± 0.30
dSPG	$\mathbf{89.42} \pm 1.95$	$\mathbf{95.43} \pm 1.07$	96.26 ± 0.48	96.43 ± 0.48
dSPGCN	87.02 ± 1.28	94.72 ± 1.33	$\mathbf{96.61} \pm 0.70$	$\mathbf{97.85} \pm 0.29$
dSPGiGCN	86.74 ± 2.27	94.63 ± 0.47	96.15 ± 0.84	97.47 ± 0.39

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TABLE IV

EXPERIMENTAL DETAILS (OA [%], AA [%], AND KAPPA [%]) ON THE AVIRIS DATASET WITH TEN LABELED SAMPLES PER CLASS

classNo.	dSPG	SPG [33]	HybridSN [51]	MDGCN [49]	DMSGer [7]	GiGCN [21]	MSSGU [36]	MSSGU⁻ [36]	CEGCN [35]
1	91.36 ± 3.98	98.86 ± 1.20	95.23 ± 4.82	93.41 ± 4.37	**99.77** ± 0.68	89.77 ± 8.93	99.32 ± 1.04	99.09 ± 1.51	96.59 ± 5.86
2	79.73 ± 6.87	71.07 ± 5.14	67.16 ± 5.58	62.40 ± 7.68	66.57 ± 7.46	74.47 ± 9.59	**80.34** ± 1.17	73.33 ± 8.68	66.92 ± 12.20
3	**82.63** ± 6.23	80.57 ± 9.01	67.34 ± 11.63	64.60 ± 9.99	72.34 ± 12.42	78.71 ± 12.50	80.95 ± 1.26	74.93 ± 13.09	68.01 ± 10.02
4	**99.69** ± 0.84	99.42 ± 1.11	96.25 ± 5.02	73.75 ± 11.45	99.55 ± 1.20	92.01 ± 4.53	96.29 ± 3.97	97.46 ± 2.83	92.54 ± 7.37
5	84.83 ± 6.39	82.40 ± 5.98	80.82 ± 5.48	86.06 ± 11.02	81.03 ± 8.72	82.87 ± 9.42	**89.30** ± 7.92	86.08 ± 10.37	86.04 ± 9.76
6	87.41 ± 6.02	93.00 ± 3.29	91.83 ± 3.90	86.24 ± 5.63	83.68 ± 12.02	89.82 ± 2.76	95.60 ± 4.59	91.29 ± 4.38	**97.91** ± 2.11
7	98.12 ± 3.02	**100** ± 0.00	**100** ± 0.00	**100** ± 0.00	**100** ± 0.00	93.13 ± 7.63	**100** ± 0.00	**100** ± 0.00	**100** ± 0.00
8	99.81 ± 0.07	99.81 ± 0.07	99.75 ± 0.63	94.47 ± 2.95	99.85 ± 0.44	97.89 ± 1.57	**99.98** ± 0.06	99.87 ± 0.25	99.48 ± 0.93
9	94.00 ± 18.97	**100** ± 0.00	**100** ± 0.00	**100** ± 0.00	**100** ± 0.00	**100** ± 0.00	**100** ± 0.00	**100** ± 0.00	**100** ± 0.00
10	**88.11** ± 6.09	83.68 ± 5.51	79.65 ± 8.81	76.67 ± 10.48	81.09 ± 5.66	80.08 ± 9.85	86.69 ± 4.68	82.38 ± 8.24	80.71 ± 14.20
11	**89.74** ± 6.60	79.77 ± 5.64	65.53 ± 7.57	69.04 ± 8.41	70.49 ± 3.28	81.49 ± 5.73	79.56 ± 7.79	82.50 ± 5.23	75.26 ± 11.61
12	**91.66** ± 9.91	78.36 ± 7.68	71.24 ± 10.44	68.15 ± 8.69	78.31 ± 7.57	79.17 ± 8.67	83.03 ± 9.06	75.18 ± 6.97	73.18 ± 12.47
13	99.01 ± 0.00	99.21 ± 0.42	97.08 ± 4.81	86.88 ± 2.41	98.86 ± 1.76	95.30 ± 4.80	99.85 ± 0.22	99.65 ± 0.50	**99.95** ± 0.15
14	**99.70** ± 0.58	99.20 ± 0.92	86.19 ± 5.82	88.18 ± 8.08	85.55 ± 7.53	95.79 ± 3.71	98.72 ± 2.07	97.78 ± 2.03	96.36 ± 3.86
15	86.11 ± 9.10	90.38 ± 12.42	81.30 ± 6.74	78.60 ± 10.74	93.87 ± 8.62	**94.86** ± 3.02	94.08 ± 9.60	88.89 ± 7.50	85.76 ± 8.66
16	98.82 ± 0.00	96.47 ± 5.68	97.05 ± 3.62	91.77 ± 5.84	**99.88** ± 0.35	98.24 ± 0.94	99.76 ± 0.47	**99.88** ± 0.35	99.65 ± 0.54
OA	**89.42** ± 1.95	84.85 ± 1.56	76.75 ± 1.37	75.25 ± 3.01	78.77 ± 1.37	84.66 ± 2.30	87.25 ± 2.16	84.87 ± 2.07	81.45 ± 2.48
AA	91.92 ± 2.05	90.76 ± 1.31	86.03 ± 1.04	82.51 ± 2.15	88.18 ± 1.62	88.98 ± 1.41	**92.72** ± 1.38	90.52 ± 1.13	88.65 ± 1.55
Kappa	**87.98** ± 2.21	82.90 ± 1.74	73.89 ± 1.48	72.14 ± 3.34	76.27 ± 1.52	82.38 ± 2.99	85.51 ± 2.43	82.79 ± 2.33	78.94 ± 2.71

TABLE V

OVERALL ACCURACY [%] OBTAINED BY DIFFERENT METHODS FOR THE OMIS DATASET (STANDARD DEVIATIONS IN PARENTHESES)

Number of labeled pixels per class	**dSPG**	SPG	SSM	SSNG	SNG	NNG	SVM	MLR
3	**78.30**	72.17	56.33	70.93	64.44	49.01	54.92	60.40
	(6.52)	(4.94)	(10.19)	(4.20)	(4.85)	(4.66)	(3.94)	(3.86)
5	**84.19**	79.15	69.71	74.41	72.30	51.63	62.73	64.69
	(2.92)	(3.78)	(5.61)	(4.33)	(4.61)	(3.64)	(3.19)	(3.08)
7	**86.03**	82.39	76.68	77.46	76.90	54.72	67.62	70.20
	(2.14)	(4.08)	(7.10)	(3.11)	(4.08)	(2.81)	(3.23)	(3.06)
10	**87.35**	85.15	78.45	80.47	81.79	58.83	71.95	74.45
	(1.15)	(2.79)	(4.83)	(1.19)	(2.50)	(1.92)	(3.23)	(1.98)
15	**88.13**	86.95	85.23	81.78	84.57	60.46	77.28	77.76
	(0.91)	(2.02)	(2.91)	(1.39)	(1.67)	(1.57)	(2.17)	(0.95)
20	**89.01**	88.41	84.96	82.42	86.50	61.72	79.42	79.28
	(0.99)	(2.35)	(3.53)	(1.35)	(1.92)	(0.97)	(2.37)	(0.89)
50	**93.61**	91.79	85.81	84.49	91.98	65.44	85.32	83.85
	(0.84)	(1.10)	(2.41)	(1.24)	(0.72)	(1.58)	(1.18)	(0.69)

TABLE VI

OVERALL ACCURACIES [%] OBTAINED WITH 5, 7, 10, AND 20 LABELED SAMPLES PER CLASS, RESPECTIVELY, FOR THE OMIS DATASET

	5/class	7/class	10/class	20/class
HybridSN [51]	74.12 ± 4.97	81.65 ± 2.97	86.86 ± 2.53	91.59 ± 2.48
3DOC-SSAN [53]	71.37 ± 2.86	76.40 ± 4.32	79.41 ± 3.82	88.44 ± 2.11
SpectralFormer [56]	55.07 ± 4.25	61.26 ± 4.64	66.96 ± 4.64	73.90 ± 1.99
MDGCN [49]	71.65 ± 4.79	75.23 ± 5.13	81.14 ± 4.09	87.76 ± 1.38
DMSGer [7]	64.19 ± 5.77	68.68 ± 2.75	75.88 ± 4.16	81.73 ± 1.73
GiGCN [21]	78.32 ± 3.35	81.05 ± 4.33	87.88 ± 1.63	90.99 ± 1.79
dSPG	**84.19** ± 2.92	86.03 ± 2.14	87.35 ± 1.15	89.01 ± 0.99
dSPGCN	82.11 ± 2.61	**86.14** ± 2.68	88.65 ± 1.14	91.60 ± 1.63
dSPGiGCN	82.91 ± 3.16	84.91 ± 3.88	**89.91** ± 2.10	**92.65** ± 1.36

more than 50 labeled samples per class, the proposed dSPGCN has stronger classification ability due to the extracted deep features. As the third method is based on dSPG, dSPGiGCN directly embeds the *within-superpixel* graph and *between-superpixel* graph into the network of GiGCN. The results of dSPGiGCN outperform GiGCN by about 2% with different number of labels, reflecting the advantages of the proposed *within-superpixel* graph and *between-superpixel* graph.

For the OMIS dataset, as shown in Table V, we can also observe that the advantages of the proposed approach are more significant when the number of labeled samples is small. Specifically, when only five labeled samples per class are used, the proposed approach achieves an OA of around 84% (which is 5% higher than that obtained by SPG, and at least 10% higher than that reported by the other competitors). Table VI shows that our dSPG-based methods not only maintain high classification performance with large labeled sample size, but also outperform the state-of-the-art methods under the condition of less than seven labeled samples per class. With five labeled samples per class, the OA obtained by dSPGiGCN was even 4.59% higher than that of GiGCN. In Table VII, dSPG achieved 5.87%, 5.95%, and 7.73% higher than GiGCN in OA, AA, and Kappa, respectively. We consider that the long-range dependencies established by (16a) can play a more prominent role with few labels.

Furthermore, in Tables IV and VII, dSPG maintains the highest accuracies in several classes, some of which are extremely difficult to identify by other methods (such as classes 3, 11, and 12 in the AVIRIS data and classes 1, 4, and 5 in the OMIS data). It is worth mentioning that the graph weights of dSPG are fixed values, while MSSGU uses a complex network structure and graph attention mechanism to learn better graph weights. After replacing the graph attention mechanism in MSSGU with fixed similarities among data, we named the method MSSGU^{-}. The experimental results show that our proposed dSPG outperformed MSSGU and MSSGU^{-} (by 2.17% and 4.55% on AVIRIS dataset, respectively, and by 8% on OMIS dataset). The above analyses demonstrate the remarkable performance of dSPG in semisupervised hyperspectral image classification.

C. Qualitative Analysis

Fig. 10 shows some of the classification maps obtained by using five labeled samples per class on the AVIRIS scene. Among the considered methods, the proposed dSPG achieved the best performance. As it can be observed, the advantages of dSPG are particularly relevant in the following two aspects.

1) First and foremost, the classification map of dSPG is visually more consistent with the ground truth. In comparison with SPG and SNG, dSPG provides more detailed boundaries. When compared with the other

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TABLE VII

EXPERIMENTAL DETAILS (OA [%], AA [%], AND KAPPA [%]) ON THE OMIS DATASET WITH FIVE LABELED SAMPLES PER CLASS

classNo.	dSPG	SPG [33]	HybridSN [51]	MDGCN [49]	DMSGer [7]	GiGCN [21]	MSSGU [36]	MSSGU˙ [36]	CEGCN [35]
1	**92.24** ± 3.74	86.97 ± 4.01	76.84 ± 12.71	66.40 ± 15.80	68.10 ± 14.64	80.22 ± 11.62	82.25 ± 11.35	74.57 ± 14.87	84.44 ± 9.39
2	**98.77** ± 0.87	97.09 ± 3.31	91.67 ± 4.03	91.64 ± 8.01	88.78 ± 8.46	93.75 ± 3.01	91.61 ± 5.51	94.15 ± 4.08	93.45 ± 4.36
3	86.07 ± 7.43	**88.96** ± 7.68	65.38 ± 15.31	63.31 ± 12.87	65.08 ± 8.91	76.90 ± 7.58	77.32 ± 16.03	72.12 ± 14.98	77.79 ± 9.38
4	**73.94** ± 2.66	63.87 ± 8.84	66.43 ± 11.37	63.75 ± 11.92	48.60 ± 11.27	68.05 ± 11.09	62.78 ± 11.31	64.24 ± 9.79	64.96 ± 10.03
5	**97.30** ± 2.38	97.17 ± 1.06	82.42 ± 11.80	80.20 ± 10.95	81.59 ± 9.78	85.86 ± 7.03	80.90 ± 13.60	87.68 ± 7.29	95.58 ± 3.93
6	83.24 ± 5.10	82.56 ± 5.83	83.37 ± 8.16	83.98 ± 12.86	65.02 ± 14.04	**87.44** ± 6.86	85.17 ± 11.37	83.72 ± 8.42	86.96 ± 5.03
7	**99.66** ± 0.40	94.28 ± 10.80	89.79 ± 13.89	96.51 ± 5.23	89.31 ± 7.12	96.32 ± 3.89	97.87 ± 1.82	98.67 ± 1.34	97.23 ± 2.24
8	**99.17** ± 0.27	98.22 ± 1.64	91.03 ± 6.23	86.50 ± 12.76	93.21 ± 5.95	94.27 ± 5.02	94.37 ± 9.22	93.16 ± 6.91	96.35 ± 3.34
OA	**84.19** ± 2.92	79.46 ± 3.32	74.12 ± 4.97	71.65 ± 4.79	64.19 ± 5.77	78.32 ± 3.35	75.71 ± 3.69	75.18 ± 4.09	78.12 ± 4.06
AA	**91.30** ± 1.54	88.64 ± 0.93	80.87 ± 2.93	79.04 ± 2.92	74.96 ± 4.41	85.35 ± 2.61	84.03 ± 2.34	83.54 ± 2.66	87.09 ± 1.75
Kappa	**80.80** ± 2.67	74.93 ± 3.56	67.94 ± 5.60	65.26 ± 5.21	57.23 ± 6.19	73.07 ± 5.32	70.31 ± 3.85	69.66 ± 4.40	73.32 ± 4.41

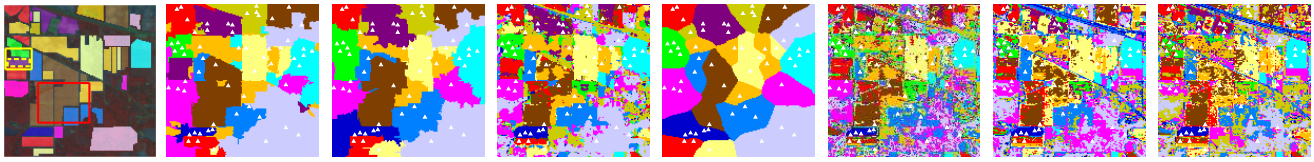

Fig. 10. Classification maps using five labeled pixels per class (illustrated in white triangles) on the AVIRIS dataset. (a) False color composition/Ground truth. (b) dSPG (proposed). (c) SPG. (d) SSNG. (e) SNG. (f) NNG. (g) SVM. (h) MLR.

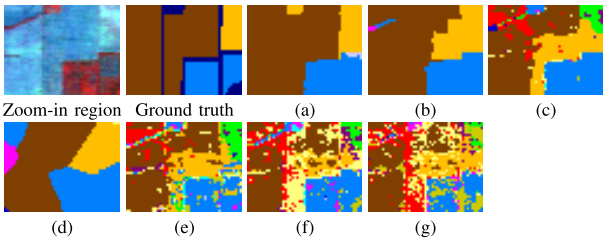

Fig. 11. Illustration of the advantages of the proposed dSPG in the classification of region boundaries. (a) dSPG (proposed). (b) SPG. (c) SSNG. (d) SNG. (e) NNG. (f) SVM. (g) MLR.

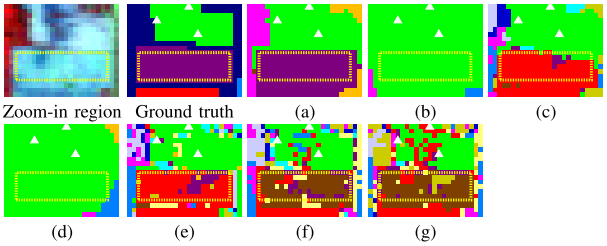

Fig. 12. Illustration of the advantages of the proposed dSPG in the classification of isolated regions with no labeled samples. (a) dSPG (proposed). (b) SPG. (c) SSNG. (d) SNG. (e) NNG. (f) SVM. (g) MLR.

methods, dSPG exhibits less classification noise. If we take the region in the red box as an example (Fig. 11), the proposed dSPG clearly obtained better boundaries than the competitors.

2) It is also interesting to observe that, in Fig. 10(b), there are some classified regions without labeled pixels. In particular, Fig. 12 provides an example focused on the region in yellow box, which is part of the *Soybean-clean* class in purple, however, without labeled samples. The proposed dSPG correctly classifies this region. This means that the proposed dSPG, by increasing the intraclass subset connection, resulted in better performance. This is a particular advantage of the proposed dSPG.

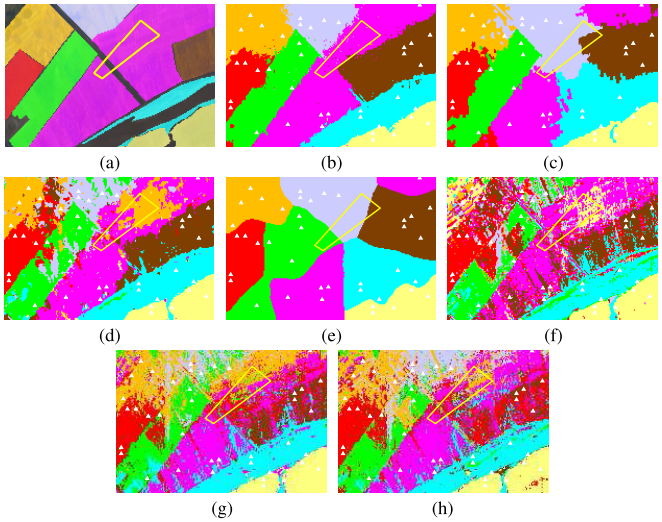

Fig. 13. Classification maps using five labeled pixels per class (marked as white triangles) for the OMIS scene. (a) False color composition/Ground truth. (b) dSPG (proposed). (c) SPG. (d) SSNG. (e) SNG. (f) NNG. (g) SVM. (h) MLR.

For the OMIS dataset, the classification maps with five labeled samples per class are depicted in Fig. 13. Among the considered methods, it is clear that the proposed dSPG achieved the best classification map. Notice that, for this dataset, each class in the ground truth is distributed as a homogeneous region with well-annotated boundaries. The proposed dSPG obtained better smoothness (inside regions) and better boundaries (across regions). Furthermore it is very interesting to note that, for the class *Pear* in magenta, as shown in the region in the yellow box, the proposed dSPG obtained a successful classification while SPG and SNG failed. For a more comprehensive analysis, we look into the distribution of the labeled samples marked by white triangles. It can be observed that there are no labeled samples in this region. Again, this is related to the successful characterization performed by the *between-superpixel* graph of the proposed dSPG.

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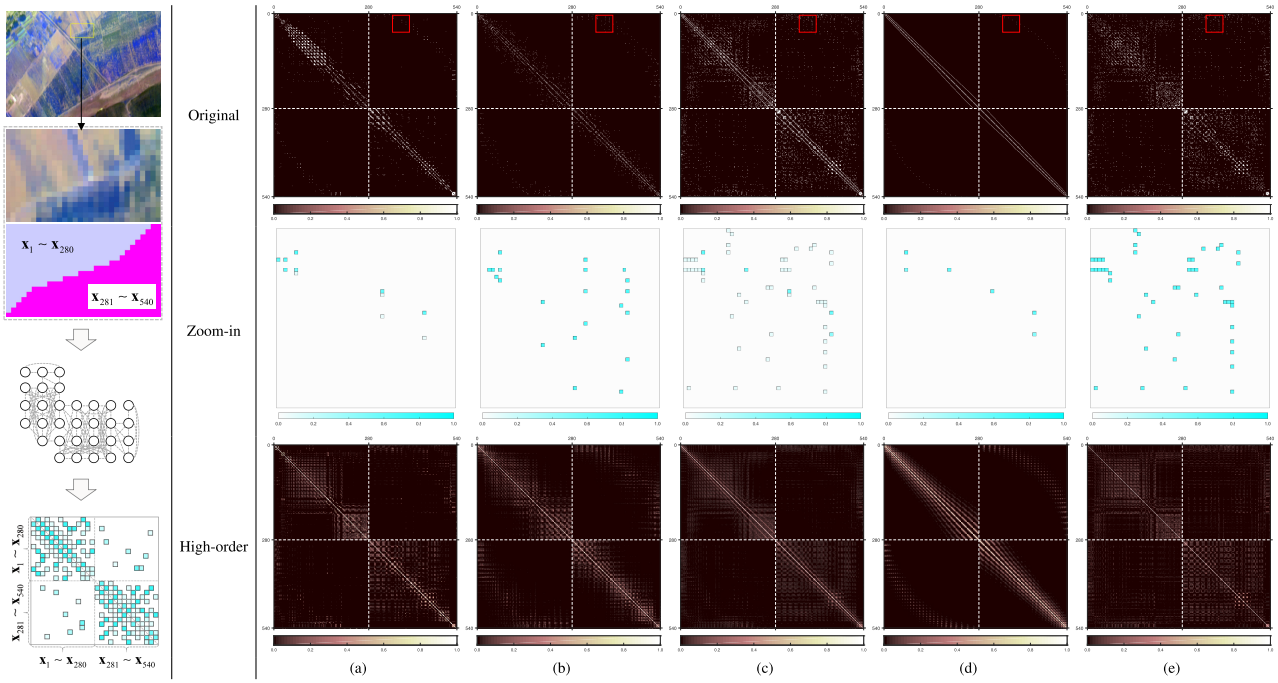

Fig. 14. Visualization of the weight matrices of dSPG and other graphs (GQ values given in the parentheses). The graph weights are denoted by the corresponding colors in the color map. (a) dSPG (proposed, 0.0307). (b) SPG (0.0936). (c) SSNG (0.0591). (d) SNG (0.0442). (e) NNG (0.1612).

D. Discussion

Here, we evaluate the proposed dSPG in terms of graph quality and the robustness to noise, respectively.

1) Graph Evaluation: In this section, we perform an evaluation of the quality of the obtained graphs. Based on Propositions 1 and 2, we propose a criterion to evaluate the quality of the graph

$$\mathrm{GQ} = \frac{\mathrm{GD}}{\mathrm{GS} + N_{\mathrm{lr}}} = \frac{\sum_{\mathbf{y}_i \neq \mathbf{y}_j} \overline{\mathrm{w}}_{i,j}}{\sum_{\mathbf{y}_i = \mathbf{y}_j} \overline{\mathrm{w}}_{i,j} + N_{\mathrm{intra}}} \tag{24}$$

where $\overline{\mathrm{w}}_{i,j}$ denotes the sparse graph weight between pixel i and pixel j; $\mathbf{y}_i \neq \mathbf{y}_j$ denotes that pixel i and pixel j have different class labels and $\mathrm{GD} = \sum_{\mathbf{y}_i \neq \mathbf{y}_j} \overline{\mathrm{w}}_{i,j}$ counts the total weights of pixels that should not be connected; $\mathbf{y}_i = \mathbf{y}_j$ denotes that pixel i and pixel j are with the same class label and $\mathrm{GS} = \sum_{\mathbf{y}_i = \mathbf{y}_j} \overline{\mathrm{w}}_{i,j}$ counts the total weights of pixels which should be connected; and N_{intra} denotes the number of intraclass connections. GQ counts the ratio between the total weights for samples belonging to different classes and to the same class. From (24) we can conclude that, the smaller the GQ, the better the graph.

Given that background pixels do not have class labels, their connections cannot be evaluated for quality. Therefore, we selected a region without background pixels in a real hyperspectral image, as shown in the first column on the left side of Fig. 14. There are two classes of ground materials in this region, *Corn* (mauve) and *Pear* (rose red), where the pixel set of *Corn* is $[\mathbf{x}_1, \ldots, \mathbf{x}_{280}]$ and the pixel set of *Pear* is $[\mathbf{x}_{281}, \ldots, \mathbf{x}_{540}]$. We analyze the quality of different graphs on this data, and present three visualizations on the right side of Fig. 14, including the original graph matrix, the enlarged subgraph composed of interclass connections, and the high-order graph matrix calculated by (21). To enlarge the difference of graphs (SNG, SSNG, and dSPG), N_{intra} adopts the number of intraclass connections with spatial distances exceeding 1.

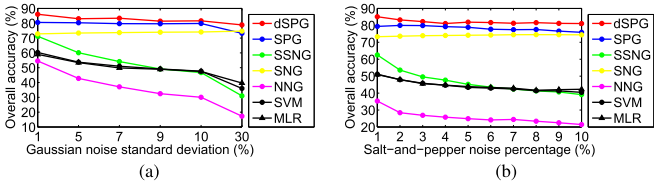

Fig. 15. Overall accuracies obtained with different noise levels on the AVIRIS dataset, using five labeled pixels per class. (a) Gaussian noise. (b) Salt-and-pepper noise.

Of the five original graphs, only dSPG, SSNG, and NNG have a large number of intraclass connections, as shown in the first line of Fig. 14. The enlarged subgraphs indicate that only dSPG and SNG have the weakest interclass connections. Besides, the high-order dependencies generated by dSPG also exhibit fewer interclass relationships and the most intraclass relationships. Thus, dSPG is the most ideal graph with the highest discriminability. This conclusion is also supported by the quantitative indicators obtained by (24) (Fig. 14, where dSPG has the smallest GQ value).

2) Effect of Noise: Since the proposed method utilizes spatial information, in this section we further evaluate the robustness of dSPG against noise. We add two kinds of noise (with different levels) to the AVIRIS dataset. The first one is Gaussian noise with zero mean. As shown in Fig. 15(a), the standard deviation of Gaussian noise (as a percentage of the gray level) ranges from 1% to 30%. The second one is salt-and-pepper noise, as shown in Fig. 15(b), with the percentage of corrupted pixels varying from 1% to 10%.

As it can be observed from Fig. 15, the proposed dSPG achieves the highest classification accuracy. However, competitors only using spectral information are severely affected

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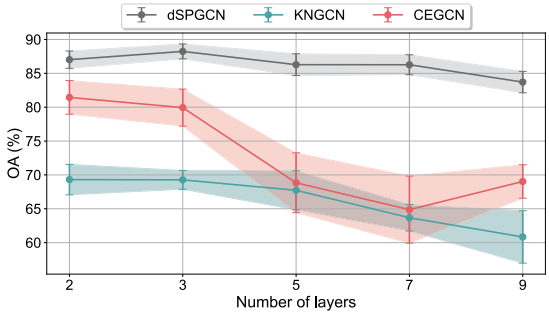

Fig. 16. Performance of dSPGCN and other GCN-based algorithms in different network depths.

by noise. Therefore, the proposed dSPG can maintain stable hyperspectral image classification performance under large levels of both Gaussian and salt-and-pepper noise.

3) Effect of Network Depth: To demonstrate the advantages of our approach in dealing with oversmoothing problems, we analyze the effect of network depth. Existing GCNs connect a large number of pixels of different classes, causing their features to converge to indistinguishable values under several layers of graph convolution, such as KNGCN and CEGCN in Fig. 16. Among them, KNGCN is a method to directly use KNG in GCN, which is also a common structure of the existing GCNs. As it can be seen from Fig. 16, our method performs better with three layers than with two layers, and maintains a high accuracy with seven layers. Although the accuracy decreases slightly at the depth of nine, it is still far superior to other GCNs. When the CEGCN and KNGCN exceed three layers, their performance deteriorates rapidly, and their stability deteriorates significantly. The above comparisons indicate that our proposed dSPGCN has the ability to resist oversmoothing.

4) Ablation Study: To analyze the role of some important components of dSPG, the following ablation experiments were performed (Table VIII).

1) *Without SpaNN:* This version removes spatial neighboring connections within and between superpixels. The dramatic performance degradation (over 20% less than dSPG) validates the importance of our proposed connections. This result shows that the graph method with only spectral information increases connections between classes and cannot solve the problem of oversmoothing in SPG. Our proposed dSPG more fully combines spatial information and spectral information to build the graph, increasing the connections between intraclass subsets while reducing the interclass connections.
2) *Without b-SpecNN:* This version removes the spectral neighboring connections between superpixels under the mutual agreement constraint, that is, the connections described in (14) and (15). This component helps dSPG capture long-range dependencies between ground materials of the same class. When dSPG removes these connections, the accuracy drops by nearly 3%.
3) *Without Representative:* When the most representative pixels and the most representative weights between superpixels are removed, a superpixel may be regarded as a node in the graph, causing oversmoothing within the superpixel. Experimental results show that the accuracy of dSPG without this component is reduced by 1%.
4) *Without Mutual Connection:* The mutual connections adopted in (15) exhibit high reliability, sparsity, and mutual agreement. This component provides a performance improvement of about 1%.
5) *Without w-SpecNN:* The w-SpecNN denotes the spectral neighboring connections within superpixels, which is used to deal with the mixing problem of superpixel in dSPG. As shown in Fig. 17, as the number of superpixels decreases, the version with w-SpecNN is significantly better than the version without w-SpecNN. This proves that w-SpecNN has more advantages in dealing with the mixing problem. Note that in order to avoid excessive connections within superpixels, we set a smaller number of neighborhoods, and as the number of superpixels increases, the number of neighborhoods decreases. Thus, the trend of the curve on the AVIRIS dataset is different from Fig. 8(a).

TABLE VIII
ABLATION STUDY ON THE AVIRIS DATASET

	OA	AA	Kappa
dSPG	85.30 ± 3.44	88.84 ± 2.63	83.37 ± 3.81
w/o SpaNN	63.88 ± 4.92	75.71 ± 2.91	59.90 ± 5.22
w/o b-SpecNN	82.58 ± 3.39	85.80 ± 2.74	80.32 ± 3.79
w/o representative	84.35 ± 3.26	88.87 ± 2.04	82.34 ± 3.61
w/o mutual-connection	84.41 ± 3.95	88.82 ± 2.94	82.42 ± 4.37

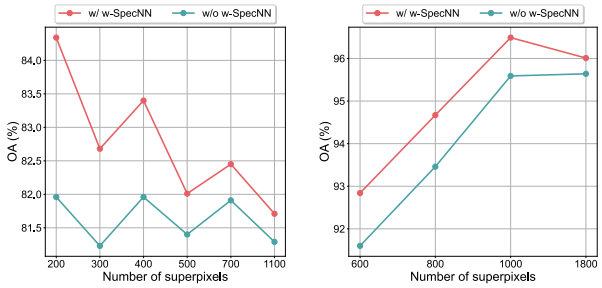

Fig. 17. Ablation study on w-SpecNN. (a) AVIRIS. (b) ZY-1(02D)-HongHu.

5) Effect of Mixed Superpixels: We propose a mixed superpixel ratio (MSR) to measure the degree of mixing within superpixels, as follows:

$$\mathrm{MSR} = \frac{\mathrm{sp}_m}{\mathrm{sp}_n} \times 100\% \tag{25}$$

where sp_m denotes the number of superpixels containing more than one class of pixels and sp_n denotes the total number of superpixels. A larger MSR value indicates a larger proportion of mixed superpixels.

In Fig. 18, as the number of superpixels decreases, the MSR value increases, and our proposed dSPG maintains higher precisions, indicating that dSPG can deal with the problem of mixing in superpixels. However, other superpixel graph methods are greatly affected by mixed superpixels, especially the GiGCN method, which deteriorates seriously when $\mathrm{sp}_n \leq 300$.

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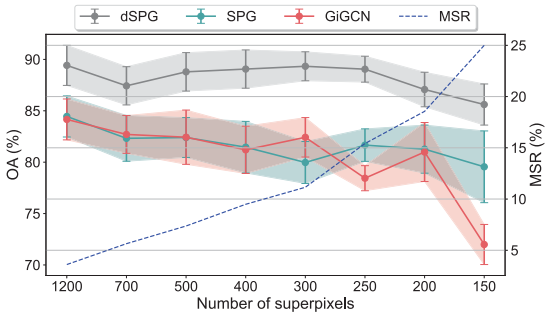

Fig. 18. Influence of mixed superpixels on dSPG.

E. Denoising Application

To remove noise in remotely sensed hyperspectral images, we adopt the following framework to implement graph filtering:

$$\widehat{\mathbf{X}} = \arg\min_{\mathbf{X}} \|\mathbf{X} - \mathbf{X}_0\|_F^2 + \mu \mathrm{tr}\left(\mathbf{X}^\top \mathbf{L}\mathbf{X}\right) = (\mathbf{I} + \mu\mathbf{L})^{-1}\mathbf{X}_0 \tag{26}$$

where $\mathbf{X}_0$ denotes the noisy image to be processed, and $\widehat{\mathbf{X}}$ denotes the filtered hyperspectral image. In our experiments, $\mathbf{X}_0$ is produced by adding Gaussian noise with standard deviation of 30% $\times$ 255 to the AVIRIS scene. Then, graph filtering based on dSPG, SSNG, and SNG is performed on the AVIRIS image by (26), considering gaussian filter (Gauss) and bilateral filter (BF) [57] as comparison methods.

As it can be seen from Fig. 19, dSPG-based graph filtering produces a smooth image with the best recovered boundaries. Other methods either oversmooth or introduce noise, thus producing worse results.

To quantitatively evaluate the effect of different denoising methods, three commonly used indices including the peak signal-to-noise ratio (PSNR), the structural similarity (SSIM), and the spectral angle mean (SAM) [58] were adopted. Besides, we classified the filtered images using the MLR with five labeled pixels per class. Then, the fisher ratio (FR) of the filtered images is calculated to measure the separability of their features. As listed in Table IX, the graph filtering results based on the proposed dSPG not only provide the best classification performance (i.e., the highest OA and FR), but also achieve the largest PSNR and SSIM (and the smallest SAM).

F. Other Applications

It should be noted that the two images considered so far (AVIRIS and OMIS) were both obtained by airborne earth observation instruments, which are the most widely used in practice. In this section, we explore more innovative applications of our newly proposed dSPG-based semisupervised classification framework by testing it with a satellite remote sensing dataset (HongHu), and a remote sensing dataset collected by an unmanned aerial vehicle (WHU-Hi).

1) Satellite Image: The classification maps obtained for the ZY-1(02D)-HongHu dataset are shown in Fig. 20, using five labeled samples per class. The obtained results reveal two advantages of the proposed dSPG.

TABLE IX
QUANTITATIVE INDICES USED TO EVALUATE DIFFERENT DENOISING METHODS ON THE AVIRIS DATASET

	Noisy AVIRIS	**dSPG**	SSNG	SNG	Gauss	BF
OA	37.59	**74.70**	43.97	50.86	66.32	71.79
FR [$\times 10^{-3}$]	0.57	**5.04**	1.26	2.18	3.31	4.26
PSNR (dB)	10.52	**18.51**	14.23	17.22	17.58	17.93
SSIM	0.22	**0.49**	0.36	0.46	0.34	0.45
SAM	31.99	**9.05**	19.39	15.70	10.68	10.28

TABLE X
CLASS NAMES OF THE HONGHU AND WHU-HI DATASETS

ZY-1(02D)-HongHuA			WHU-Hi		
Class	Name	Pixels	Class	Name	Pixels
C1	Vegetation	682	C1	Bare soil	2240
C2	Pond	574	C2	Chinese cabbage	10560
C3	Impervious surface	471	C3	Pakchoi	3230
C4	River	3446	C4	Cabbage	5027
C5	Cropland	1634	C5	Tuber mustard	594
			C6	Brassica parachinensis	2584
			C7	Brassica chinensis	4141
			C8	Small brassica chinensis	858
			C9	Lactuca sativa	1648
			C10	Celtuce	1002
			C11	White radish	1349
			C12	Garlic sprout	936

1) First, dSPG preserves detailed boundaries that are most consistent with the distribution of land cover. For example, the proposed dSPG clearly delineates the boundaries of the Yangtze River. However, NNG, SVM, and MLR confuse the *River* from the *Pond*, and SPG, SVM, and SNG confuse the *River* from the *Cropland*. For the region in the white box, dSPG and SSNG can get the best segmentation of *Pond*, while other graph-based methods fail. Furthermore, it is interesting to observe that only the dSPG, SPG, and SSNG ensure the continuity of ponds in the blue box region. The results provided by other methods are quite different from the real scene.
2) Second, dSPG can obtain better classification results, when there are no labeled pixels in the region, given the regions in black boxes as an example. However, in one of the *Pond* regions, SPG, SNG, SVM, and MLR unreasonably predict a large number of different land cover classes, including *Vegetation*, *Cropland*, and *Impervious surface*. This advantage is also evident in another two *Cropland* regions. Globally, NNG, SVM, and MLR all produce more classification noise.

From the obtained classification maps, we can conclude that the proposed dSPG achieves the best results. The detailed indices including OA, average accuracy (AA), and Kappa coefficient under 10 Monte Carlo runs are reported in Table XI. The proposed dSPG achieved the highest scores, and the proposed dSPGCN also surpassed other state-of-the-art deep learning methods, demonstrating their superiority on this dataset. In Table XII, we further compared the proposed dSPG and deep learning methods, with ten labeled pixels per class. Our method is still superior to other competitors. Since our work focused on the semisupervised classification of hyperspectral images with few labels, we did not carry out experiments with more labeled pixels.

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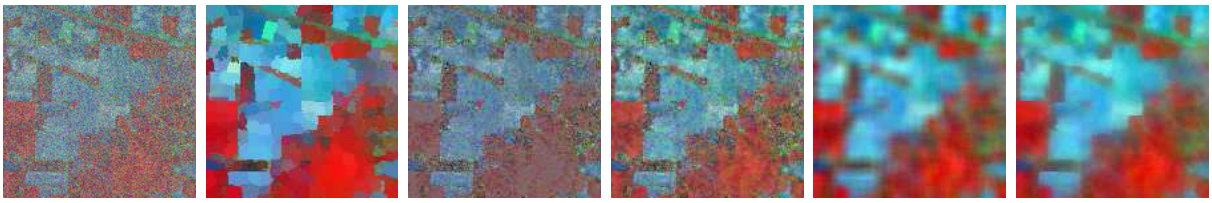

(a) (b) (c) (d) (e) (f)

Fig. 19. False color composition of the AVIRIS dataset before and after denoising. (a) Noisy AVIRIS scene. (b) dSPG (proposed). (c) SSNG. (d) SNG. (e) Gauss. (f) BF.

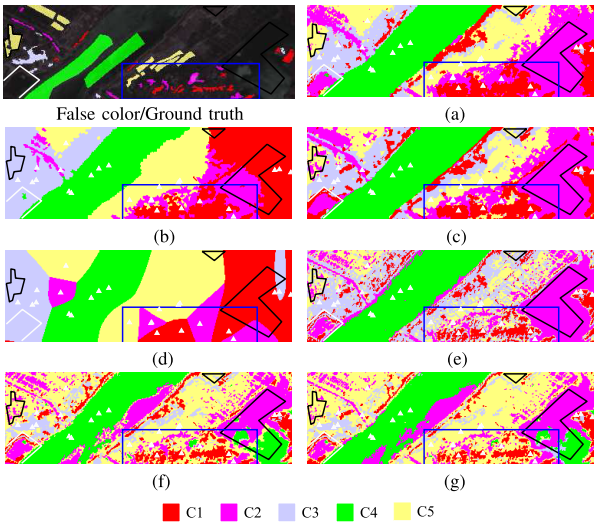

Fig. 20. Classification maps using five labeled pixels per class (marked as white triangles) on the ZY-1(02D)-HongHu satellite dataset. (a) dSPG (proposed). (b) SPG. (c) SSNG. (d) SNG. (e) NNG. (f) SVM. (g) MLR.

TABLE XI

COMPARISON OF OA [%], AA [%], AND KAPPA [%] FOR THE ZY-1(02D)-HONGHU DATASET WITH FIVE LABELED PIXELS PER CLASS

	OA	AA	Kappa
dSPG	$\mathbf{97.76 \pm 1.04}$	$\mathbf{95.84 \pm 2.80}$	$\mathbf{96.62 \pm 1.57}$
dSPGCN	94.75 ± 2.77	94.04 ± 1.29	92.26 ± 3.93
SPG	91.19 ± 1.04	86.91 ± 2.64	86.84 ± 2.32
SSM	84.50 ± 3.78	79.27 ± 5.51	77.14 ± 5.52
SSNG	95.06 ± 1.22	94.76 ± 0.92	92.64 ± 1.79
SNG	78.63 ± 2.94	73.44 ± 2.18	69.17 ± 3.74
NNG	90.35 ± 1.78	89.35 ± 1.70	85.72 ± 2.56
MLR	90.78 ± 3.38	90.05 ± 2.16	86.45 ± 4.71
SVM	84.60 ± 2.76	88.71 ± 1.88	78.22 ± 3.80
HybridSN [51]	89.15 ± 4.40	82.59 ± 4.69	83.93 ± 6.27
3DOC-SSAN [53]	85.39 ± 5.08	78.23 ± 4.88	78.32 ± 7.11
SpectralFormer [56]	87.20 ± 4.91	79.49 ± 4.21	80.84 ± 7.01
MDGCN [49]	88.90 ± 4.22	90.62 ± 2.77	84.06 ± 5.59
DMSGer [7]	84.07 ± 2.21	78.54 ± 2.26	76.55 ± 3.09
GiGCN [21]	92.81 ± 3.95	93.66 ± 2.28	89.56 ± 5.45
MSSGU [36]	89.51 ± 3.72	90.87 ± 2.64	84.87 ± 5.22
MSSGU$^-$ [36]	91.55 ± 4.20	90.53 ± 2.40	87.69 ± 5.77
CEGCN [35]	93.69 ± 2.85	91.12 ± 2.98	90.61 ± 4.15

2) Unmanned Aerial Vehicle Image: In Fig. 21, the classification maps provided by SVM and MLR exhibit severe classification noise. While the proposed dSPG not only obtains smoother classification results (as shown in the black box), but also preserves boundaries (as shown in the blue box), which are difficult to achieve by other methods.

As shown in Table XIII, the methods of the proposed dSPG family obtained the best OA, AA, and Kappa on the WHU-Hi dataset.

TABLE XII

COMPARISON OF OA [%], AA [%], AND KAPPA [%] FOR THE ZY-1(02D)-HONGHU DATASET WITH TEN LABELED PIXELS PER CLASS

	OA	AA	Kappa
dSPG	$\mathbf{98.74 \pm 0.42}$	$\mathbf{98.07 \pm 0.92}$	$\mathbf{98.09 \pm 0.63}$
HybridSN [51]	87.74 ± 7.67	88.62 ± 3.19	82.67 ± 10.00
MDGCN [49]	88.02 ± 3.33	91.80 ± 2.05	83.02 ± 4.48
DMSGer [7]	79.22 ± 7.07	75.31 ± 5.42	70.49 ± 9.14
GiGCN [21]	93.64 ± 1.87	95.35 ± 1.33	90.70 ± 2.67
MSSGU [36]	93.17 ± 1.30	94.00 ± 1.30	89.97 ± 1.85
MSSGU$^-$ [36]	92.20 ± 1.80	92.96 ± 1.65	88.59 ± 2.55
CEGCN [35]	96.74 ± 0.97	96.24 ± 1.10	95.12 ± 1.43

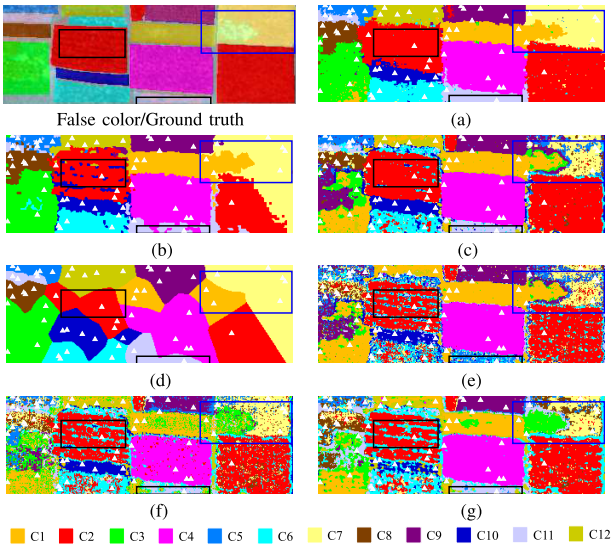

Fig. 21. Classification maps using five labeled pixels per class (marked as white triangles) on the WHU-Hi unmanned aerial vehicle dataset. (a) dSPG (proposed). (b) SPG. (c) SSNG. (d) SNG. (e) NNG. (f) SVM. (g) MLR.

From these experiments, we can conclude that the proposed dSPG method offers significant advantages in the classification of remotely sensed hyperspectral data collected by airborne, satellite, and unmanned aerial instruments. In addition, dSPG can be applied for both spatially fragmented and fine-grained land cover classification, covering the practical totality of remote sensing applications. Compared with other competitors, our proposed dSPG exhibits stronger generalization ability.

Besides the classification and denoising tasks of hyperspectral data, dSPG, as a graph of image, can also be used in scene classification, semantic segmentation, change detection, unmixing, and super-resolution of remote sensing image [59], [60]. Some of these applications are integrated into the framework of a foundation model called SpectralGPT [59]. Inspired by SpectralGPT, dSPG can be used to build the graph autoencoder [61] conveniently in the form of graph

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TABLE XIII
COMPARISON OF OA [%], AA [%], AND KAPPA [%] FOR THE WHU-Hi DATASET, WITH FIVE LABELED PIXELS PER CLASS

	OA	AA	Kappa
dSPG	89.07 ± 1.58	$\mathbf{89.94 \pm 0.91}$	87.19 ± 1.81
dSPGCN	$\mathbf{89.15 \pm 2.74}$	89.53 ± 1.65	$\mathbf{87.24 \pm 3.14}$
SPG	82.87 ± 2.06	89.18 ± 0.72	80.31 ± 2.27
SSM	76.61 ± 1.51	65.17 ± 2.27	72.56 ± 1.77
SSNG	76.68 ± 1.66	77.37 ± 1.51	72.85 ± 1.88
SNG	74.62 ± 1.85	82.43 ± 1.70	71.04 ± 2.01
NNG	63.01 ± 1.72	58.18 ± 1.92	57.26 ± 1.91
MLR	65.59 ± 3.42	61.72 ± 1.39	60.20 ± 3.59
SVM	65.68 ± 4.05	63.39 ± 2.56	60.35 ± 4.31
HybridSN [51]	73.56 ± 2.01	69.85 ± 2.93	68.79 ± 2.38
3DOC-SSAN [53]	71.59 ± 3.74	69.44 ± 3.94	66.89 ± 4.28
SpectralFormer [56]	58.33 ± 3.92	56.58 ± 3.74	52.24 ± 4.09
MDGCN [49]	75.36 ± 3.32	75.24 ± 2.04	71.29 ± 3.62
DMSGer [7]	76.37 ± 3.39	81.79 ± 2.43	72.93 ± 3.63
GiGCN [21]	87.54 ± 2.33	88.57 ± 1.10	85.37 ± 2.62
MSSGU [36]	83.81 ± 3.98	84.40 ± 2.43	81.02 ± 4.37
MSSGU$^-$ [36]	80.04 ± 3.00	79.20 ± 2.18	76.53 ± 3.18
CEGCN [35]	85.45 ± 3.45	88.06 ± 1.92	83.06 ± 3.78

regularization or graph convolutional layer. Taking advantage of dSPG's good ability to model data relationships, the graph autoencoder based on dSPG will generate latent features under better manifold assumptions than the autoencoder in SpectralGPT. Thus, dSPG can also be used to build a foundation model that serves a variety of downstream tasks.

V. CONCLUSION AND FUTURE LINES

In this article, we have introduced a new dSPG comprising a *within-superpixel* graph and a *between-superpixel* graph for semisupervised hyperspectral image classification. The proposed *within-superpixel* graph disconnects pixels belonging to different classes in the same superpixel so as to reduce the interclass connections weights, while the *between-superpixel* graph connects spectral adjacent superpixels aiming at increasing intraclass subset connections. Our experimental results, conducted with a variety of hyperspectral images, demonstrate that our newly proposed dSPG method can preserve class boundaries and homogeneity, outperforming several state-of-the-art approaches. As a regularizer, dSPG can be widely used in the tasks of denoising, clustering, and domain adaptation of hyperspectral images. By transferring the deep features of dSPGCN or modifying the network structure of dSPGCN, our method can also be applied to object detection, segmentation, anomaly detection, and image reconstruction.

However, our current work did not focus on reducing the computational complexity. In dSPG, the calculation of complex connections within and between superpixels and the operation of traversing pixels in dSPG may take more time than the existing graphs. In the future, we will alleviate the problem by reducing the graph size. Besides, we will apply the graph attention mechanism to update the weights instead of using fixed weights.

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Long Yu (Graduate Student Member, IEEE) received the B.S. and M.S. degrees from the School of Automation Science and Engineering, South China University of Technology, Guangzhou, China, in 2015 and 2019, respectively. He is currently pursuing the Ph.D. degree with the School of Geography and Planning, Sun Yat-sen University, Guangzhou.

His research interests include image processing and machine learning on semisupervised classification and domain adaptation.

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Jun Li (Fellow, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2004, the M.E. degree in remote sensing from Peking University, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from the Instituto de Telecomunicações, Instituto Superior Técnico (IST), Universidade Técnica de Lisbon, Lisbon, Portugal, in 2011.

From 2013 to 2021, she was a Full Professor with Sun Yat-sen University, Guangzhou, China. Since 2022, she has been a Full Professor with China University of Geosciences, Wuhan, China. She has obtained several prestigious funding grants at the national and international levels. She has authored or co-authored more than 160 journal citation report (JCR) papers, 60 international conference papers, and a book chapter. Her main research interests comprise machine learning and its applications, such as image analysis, signal processing, supervised/semisupervised learning, and active learning.

Prof. Li is the Editor-in-Chief of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. She has been a Guest Editor of several journals, including PROCEEDINGS OF THE IEEE and *ISPRS Journal of Photogrammetry and Remote Sensing*.

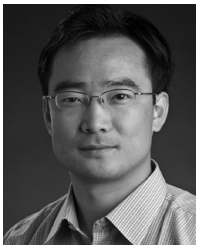
Lin He (Member, IEEE) received the B.S. degree from Xi'an Institute of Technology, Xi'an, China, in 1995, the M.S. degree from Chongqing University, Chongqing, China, in 2003, and the Ph.D. degree in pattern recognition and intelligent systems from Northwestern Polytechnical University, Xi'an, in 2007.

Since 2007, he has been with the School of Automation Science and Engineering, South China University of Technology, Guangzhou, China, where he is currently a Full Professor. His research interests include image processing, spectral–spatial–temporal information fusion, machine learning, and high-dimensional signal processing.

Antonio Plaza (Fellow, IEEE) received the M.Sc. and Ph.D. degrees in computer engineering from the Department of Technology of Computers and Communications, Hyperspectral Computing Laboratory, University of Extremadura, Cáceres, Spain, in 1999 and 2002, respectively.

He is currently a Full Professor and the Head of the Department of Technology of Computers and Communications, Hyperspectral Computing Laboratory, University of Extremadura. He has authored or co-authored more than 600 publications, including 284 journal citation report (JCR) journal papers (more than 170 in IEEE journals), 24 book chapters, and over 300 peer-reviewed conference proceeding papers. His main research interests comprise hyperspectral data processing and parallel computing of remote sensing data.

Dr. Plaza was a member of the Editorial Board of IEEE GEOSCIENCE AND REMOTE SENSING NEWSLETTER from 2011 to 2012 and *IEEE Geoscience and Remote Sensing Magazine* in 2013. He was also a member of the Steering Committee of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS). He was a recipient of the recognition of Best reviewer of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2009 and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2010, for which he has served as an Associate Editor from 2007 to 2012. He was a recipient of the Best Column Award of the *IEEE Signal Processing Magazine* in 2015, the 2013 Best Paper Award of the JSTARS journal, the most highly cited paper of *Journal of Parallel and Distributed Computing* from 2005 to 2010, and the best paper awards at the IEEE International Conference on Space Technology and the IEEE Symposium on Signal Processing and Information Technology. He was included in the 2018 and 2019 Highly Cited Researchers List (Clarivate Analytics). He has served as the Director of Education Activities for IEEE GEOSCIENCE AND REMOTE SENSING SOCIETY (GRSS) from 2011 to 2012 and the President of the Spanish Chapter of IEEE GRSS from 2012 to 2016. He has reviewed more than 500 manuscripts for over 50 different journals. He has served as the Editor-in-Chief for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING journal from 2013 to 2017. He has guest-edited ten special issues on hyperspectral remote sensing for different journals. He is an Associate Editor for IEEE ACCESS. Additional information: http://www.umbc.edu/rssipl/people/aplaza.



Lizhe Wang (Fellow, IEEE) received the B.E. and M.E. degrees from Tsinghua University, Beijing, China, in 1998 and 2001, respectively, and the Dr.Eng. degree (magna cum laude) from the University of Karlsruhe, Karlsruhe, Germany, in 2008.

He is currently the ChuTian Chair Professor with the School of Computer Science, China University of Geosciences, Beijing, and also a Professor with the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing. His research interests include high-performance computing, eScience, and remote sensing image processing.

Dr. Wang is a fellow of the IET and the British Computer Society. He serves as an Associate Editor for IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS, IEEE TRANSACTIONS ON CLOUD COMPUTING, and IEEE TRANSACTIONS ON SUSTAINABLE COMPUTING.

Zhonghui Tang is a Professor with Zhongshan School of Medicine, Sun Yat-sen University, Guangzhou, China, specializing in computational biology. He has been studying computational biology for the last ten years, with broad expertise in areas including functional genomics, single-cell omics, sequencing, and machine learning. He currently has active research programs in human genetics and functional genomics, to interpret human genomic data in the context of the human 3-D genome.

Li Zhuo received the Ph.D. degree in physical geography from Beijing Normal University, Beijing, China, in 2005.

Since September 2005, she has been with the School of Geography Science and Planning, Sun Yat-sen University, Guangzhou, China, where she is currently a Professor. Her research interests include the information extraction methods of spatiotemporal geographic data like remote sensing and geographic information, and its application on regional sustainable development topics, such as (but not limited to) surface element extraction, population distribution and movement analysis, health and risk research, and spatiotemporal data information mining method.

Dr. Zhuo is a member of the Theory and Method Specialized Committee of Chinese Geographical Information System Association, the International Association of Chinese Professionals in Geographic Information Sciences (CPGIS), the Risk Analysis Council of China Association for Disaster Prevention, and the Geographical Society of China.

Yuchen Yuan received the M.E. degree in bioengineering from South China University of Technology, Guangzhou, China, in 2020. He is currently pursuing the Ph.D. degree with Zhongshan School of Medicine, Sun Yat-sen University, Guangzhou.

His research interests include bioinformatics, single cell, and spatial omics.

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